

Inflation Tax: Crony Capitalism's Malignant Tumor

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Abstract

Money isn't supposed to grow on trees, but it does. The process of involuntary redistribution of purchasing power (effectively taxation) through secular inflation, is *the only* involuntary, non-governmental tax accepted by society, obscured by the complexity of finance. The capitalist system is successful because it's powered by decentralized, voluntary decisions. Forced redistributions are limited to market failure edge cases, to protect public goods when externalities cannot be internalized. Only the government has taxation authority, and thus should ultimately be answerable to the people through representation (no taxation without representation!). Inflation tax is the single exception. I argue that this has far reaching implications, and in this essay I lay out the case against monetary expansion, at any level, arguing that it does not lead to productive growth, but to unhealthy, unnatural, cancerous growth. After describing the inflation tax mechanism, I broadly explore many potential downstream consequences, each framed as a market distortion, leaving fully dedicated in-depth analyses of each process for future work.

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1 Inflation Tax Mechanism

When money is created *ex nihilo*, there is no financial alchemical process to credit for the creation of value out of nothing. Money creation is always a *redistribution* of purchasing power, from money holders (as the purchasing power of their money diminishes proportionally) to money printers^{1 2}. If prices adjust quickly, purchasing power redistribution is obvious, but even when prices adjust slowly, purchasing power redistribution is still reflected in disequilibrium quantity shortages. Yes, money creation can be used to finance productive endeavors, which in turn genuinely “create” value in a sense (over time and real through effort) by more efficiently reorganizing available resources, but that is separate from the act of money creation itself. One can argue that it is beneficial or even necessary to redistribute purchasing power from the inflation tax base to the inflation tax collector in order to finance productive endeavors, but this argument needs to be defended explicitly rather than making it seem as if money creation is benign value creation rather than redistribution. The three key questions to address in order to understand the mechanics of inflation are: From whom is the tax collected, who collects the tax, and on what are the taxes spent?

1.1 Tax Base

The tax base are asset holders from which inflation drains purchasing power. To identify this group, there are three relevant categories of assets: money (as defined in Kofman (2026, pp. 1–12): deposits, cash, internal ledger credits), debt (including treasuries, repo assets, and MMFs), and other assets. Other assets go up in value as a result of inflation. Debt’s value does not change if inflation is expected, since it’s priced into contract terms. Yielding deposits could be framed as loans to banks, with inflation priced into the yield, and would thus not change in value with inflation, but since they are money, it makes more sense to model the offsetting yield as inflation tax outlays (subsection 1.3) rather than a “pricing in inflation” effect. The larger share money takes up in an individual’s portfolio, the greater their contribution is to the inflation tax base.

Generally speaking, lower wealth percentiles have greater needs for liquidity as a

¹The inflation tax mechanism described in this essay rests on the definitions of money and inflation developed in Kofman (2026), a quantity-theory-of-money-like understanding: money supply simply/directly/mechanically influences the general secular price level, including financial assets.

²Money creation will be referred to as “printing” even though this will refer mostly to digital ledger entries, not actual physical printing.

portion of total assets, relative to higher percentiles ³. Lower income households need to spend a higher portion of their income on basic necessities / non-durable goods. Higher income households have a lower propensity to consume; they can afford to invest a higher portion of their assets. Higher propensities to consume require a higher portion of assets to be held in money because it's inconvenient and riskier to have to sell an asset every time before being able to spend.

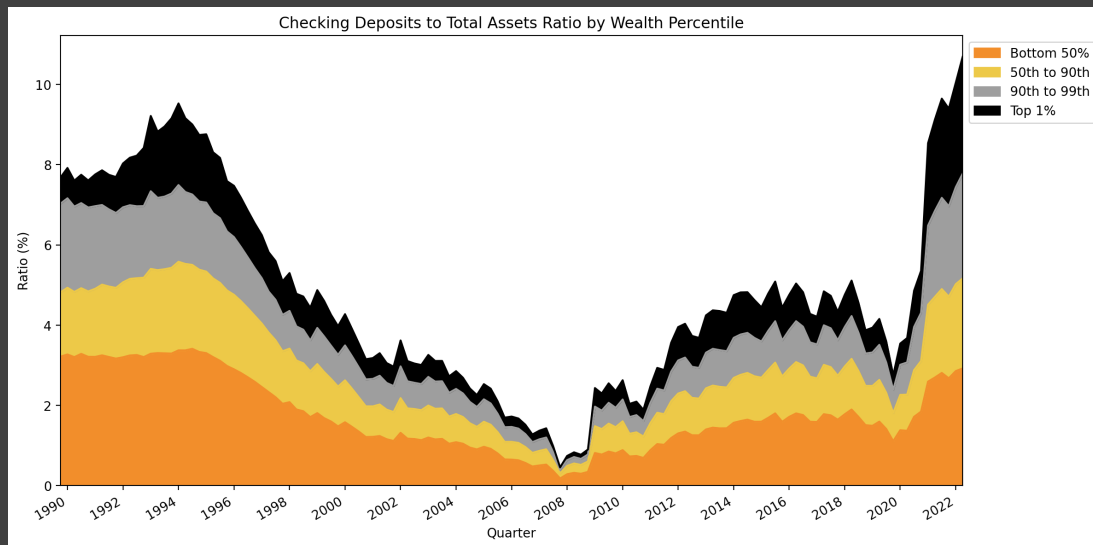


Figure 1. Checking Deposits to Total Assets Ratio by Wealth Percentile.

Series is plotted for all years for which there is available data ^a. Checking accounts include yield-bearing-Interest Checking/High-Yield Checking/Rewards Checking/NOW accounts, and other interest bearing checking accounts, which upper percentiles have greater access to for similar reasons to those explained in 1.3.1 (yield benefits and reduced frictions are only realized with large balances). The apparent change in “highest liquidity” money to asset preferences during pandemic years is corrupted by extremely rapid monetary expansion and loosened regulation distinguishing material differences between checking and savings accounts. Sources used for figure: [Board of Governors of the Federal Reserve System \(2026i,g,h,j,m,k,l,n,c,d\)](#)

^aI will focus on the U.S. economy for simplicity, but the same arguments apply to any similar fiat system.

But if banks expand deposits at roughly 6.54% a year (at an average CAGR from Q4 1946 to Q4 2024 [Board of Governors of the Federal Reserve System, 2024, 2026f,e](#)), and money is a small portion of individual's total asset portfolio, then it's not immediately clear why purchasing power transfers through money creation/devaluation

³For simplicity, I will use a discrete binary division between upper wealth percentiles and lower wealth percentiles, although of course in reality it is a more complex continuous gradient. Directionally, the core logic still applies with a more realistic breakdown. The shortcut is not meant to imply a Marxist vision of the economy as a strict division between the “bourgeoisie” and the “proletariat”, but rather to be a method of simply describing divergent directional impacts based on wealth.

are a major concern, assuming income properly adjusts. The significance becomes clearer when understanding the collector and their outlays.

1.2 Tax Collector

In many discussions of inflation tax, the collector is often framed as the government, which is not entirely the correct framing. While it's true the government enables inflation by expanding the Fed's balance sheet as it purchases treasury bonds with newly created reserves, and in doing so benefits by paying for its expenses without increasing tax income, reserves are not money for the nonbank market (Kofman, 2026, p. 3). An increase in reserves does *enable* the creation of bank deposits, which is money for the much larger and more relevant nonbank market/economy, but it's not money creation in the most relevant context (the nonbank market). Money creation is the source of upwards inflation pressure, and so the true tax collectors, the sources of money creation and inflation, are private banks.

Banks create deposits when they originate loans, but any single loan is only a temporary expansion in money supply. When a deposit is created in this way, it's spent by the borrower and the liability is passed on to other banks. Eventually, the loan must be repaid using deposits greater than the amount of deposits initially created (principal + interest). If the required deposits are obtained by the borrower from an external source with an external bank, then when the loan is repaid to the borrower's bank, the borrower's bank doesn't hold the external bank's deposit liability. The transaction is settled in reserves, and the external bank's deposit liability is destroyed. Similarly, even if all market participants use the same bank, the bank doesn't hold its own liability when the loan is repaid, the liability is destroyed. In the marginal case of this loan, holding all else equal and assuming when the borrower spent his deposits after receiving the loan that the banks those deposits were sent to had balanced incoming and outgoing payments with the borrower's bank, and when the borrower repaid the loan, the bank from which he received the principal and interest to repay the loan had balanced incoming and outgoing payments with the borrower's bank as well, then the borrower's bank initially lost reserves equal to the loan principal, but upon repayment received reserves equal to principal + interest (loosening these balanced payment requirements, or changing the interest repayment timing, does not change the fundamental outcome of this analysis as the net outcome is the same on reserve claims between banks, which cancel each other out during netting). At this point the borrower's bank has returned to its initial position, but

with increased equity equal to the total interest payment. The bank has outlay commitments to nonbank agents, which it pays out equal to its interest revenue, shifting the bank's equity liability (by the amount of interest revenue) to deposit liabilities (a small portion of the equity liability never released into the economy as a deposit liability in order to fulfill capital requirements and to maintain operational health). Whether or not this actually results in a persistent (all outside variables being equal) increase in deposits within the economy depends on four possible cases for how the interest payment was obtained (deposits initially expanded by the principal amount but were destroyed upon repayment).

1. Interest was obtained from existing deposits: system wide deposits remain unchanged.
2. Interest obtained from deposits created by an existing asset sale from a nonbank agent to a bank: system wide deposits have increased by the interest amount.
3. Interest obtained from deposits created from a new loan: system wide deposits have increased by the interest amount.
 - (a) This loan could be securitized, resulting in a persistent increase in securities (some of which are tracked in bank T-accounts but some of which are sold to nonbanks), rather than an outstanding loan increase on bank T-accounts.
4. The treasury issues new bonds which are ultimately paid for with reserves (not bank deposits), then spends those reserves back into the economy, forcing banks to create new deposits in the process, which are used to pay the aforementioned interest, and never pays down this new nominal debt level (although its inflation adjusted burden can decrease).

Options three and four require an ever-expanding pool of deposits / loans, since the isolated effect of the complete lifecycle of any single bank originated loan, is the destruction of more deposits than were created (principal created on origination, principal plus interest destroyed upon repayment). While the bank recreates most of the net destroyed deposits (equal to interest payment amounts) from the loan on its outlays, those deposits may not immediately/fully flow back into the sources from which interest payments were collected by the borrower. This means that while loan origination creates deposits originally, the full lifecycle of a loan can leave behind

residual liquidity thirst from sectors that do not receive adequate residual liquidity from sectors to which bank outflows go, a portion of which must be fulfilled by new loans/deposits, encouraging interest payment options three and four.

In this way banks create long term, persistent marginal demand for new deposits, by redistributing interest payments from borrowers to bank outlay recipients. That marginal demand is met by other banks, or the same bank, creating new deposits through loans, which will eventually also lead to more marginal demand for deposits, an endless cycle. This can help explain why deposits secularly expand so steadily, even during recessions. As the long term persistent marginal recipients of new deposits (interest revenue), sources of new deposit demand (interest payments), and deposit suppliers through loan origination, bank outlay recipients are the beneficiaries of persistent secular deposit expansion, leading to inflation. Thus private banks are the inflation tax collectors.

This can be restated much simpler. Private banks create money for nonbanks; their core business model depends on the profits from lending, which is money creation, which causes inflation. So then private banks are the regressive tax collectors, as the direct beneficiaries and causes of inflation. Of course borrowers benefit from access to financing, which is critical for the economy, but borrowers do not keep the money that was created. If there is persistent deposit creation, and thus inflation, then banks are persistently collecting inflation tax. They are the direct beneficiaries of inflation's purchasing power redistribution, not the borrowers, who benefit from the separate financial intermediation effect which does not require persistent inflation (existing deposits can be used to make interest payments, if they're circulated efficiently enough, rather than creating new deposits to make interest payments. A flexible fractional reserve architecture which allows for temporary, adaptive monetary expansion and contraction can still operate with a natural deposit creation limit: a fixed supply of bank reserves).

1.3 Outflows

In Q4 2024 , the average bank's revenue was spent on ([Hoffman, 2025](#), p. 9):

- Costs of funding / interest: 36%
 - Deposits: 29% (of total)
 - Wholesale/Other: 7% (of total)

- Operating expenses: 37%
 - Salaries/wages (Q4): 17% (of total)
 - Other operating expense (Q4): 20% (of total)
- Tax: 4%
- Credit loss provisions: 6%
- Retained earnings: 6%
- Dividends: 11%

1.3.1 Deposits

Deposit interest payments flow into traditional savings accounts, High Yield Savings Accounts (HYSA), Money Market Deposit Accounts, and Certificates of Deposit (CD) (along with high yield checking accounts and other products). CD's conflict with lower percentiles's higher liquidity needs, MMDA's have restrictive minimum deposit requirements, and HYSA are often online only banks, meaning delayed 1-3 day ACH transfers to brick and mortar checking (some HYSA banks do not offer their own checking accounts). Access to cash and other in person services are not possible with an online only HYSA and checking setup. Savings of lower percentiles are typically small relative to income, which make benefits of higher yields less attractive relative to operational costs, meaning traditional savings accounts are often a rational alternative to HYSA for many. The fixed convenience costs of switching to HYSA vs the scaling-with-deposit-size yield benefits tradeoffs make less sense the less savings an individual has. Thus there are structural expected-utility-maximizing reasons which restrict lower percentiles from benefiting from high beta savings instruments (deposits with high passthrough from short term rates to deposit yields), resulting in bank deposit outlay flows skewed towards wealthier percentiles, even proportionally to net worth (Of course upper percentiles will have larger raw flows simply because they have more money in the bank, which is not the issue being highlighted).

It's true that during the ZIRP era, when rates collapsed and the yield curve was flat, there was little distinction between high and low beta deposits. Bank deposit outlay differences were muted, and the Net Interest Income profit model was suppressed. However ZIRP was the exception, not the rule. From 1962 - 2018, the average yield curve slope (defined as 10y-3m U.S. treasury term spread) has been 133

bps (Kopp and Williams, 2018), and the average fed funds rate from July 1954 to January 2026 has been 4.60% (Board of Governors of the Federal Reserve System, 2026b). These conditions are generally conducive to bank profits (Borio et al., 2015), enabling maturity transformation. Higher rate levels mean low beta characteristics of certain types of deposits (checking/traditional savings) are amplified, allowing banks to fund themselves cheaply while their loans and securities bring in high yields. A steeper yield curve also magnifies the spread between short term deposit liability rates and long term asset yields. The widening spread between low beta deposits and bank assets corresponds to a widening spread between low beta deposits and high beta deposits, which amplifies the regressive nature of the inflation tax.

1.3.2 Wholesale Funding

Bank's wholesale funding costs flow into other banks and various shadow banking entities like insurance funds, pension funds, corporate treasuries, sovereign wealth funds, Hedge funds, central banks, broker-dealers, etc. Sovereign wealth funds and pension funds distribute these profits to the wider public, but they're typically relatively small players in the wholesale lending space (Kahn and Olson, 2021, p. 5). Central banks are only large players in times of crisis, and use profits to pay private bank's interest on reserve balances. MMFs are large players, but do not have meaningful retail funding / low percentile beneficiaries because of restrictive deposit minimums, fees, and withdrawal freeze risk that make small balances impractical, similar to other high beta deposit instruments. Other institutional players are generally private financial businesses disproportionately owned by upper percentiles (Federal Reserve Bank of St. Louis, 2025).

1.3.3 Salaries/Wages

These outflows are clearly more evenly dispersed across the percentiles than wholesale funding, and potentially deposits, as banks employ a diverse range of employees. However it's still flowing into only a very specific sector of the economy.

1.3.4 Other Operating Expenses

This is the widest dispersion of revenue, including: landlords, utility companies, physical security teams, general maintenance, data center infrastructure maintainers, software companies (cybersecurity, cloud, payment processing, data storage/analytics),

law firms, advertisers, federal regulatory programs, etc. Yet this is still not broad public good spending like typical government expenditure, for example social security, national defense, education, energy/environment, transportation, scientific discovery, etc. Governmental actors are the only entities society allows to involuntarily redistribute wealth, in large part due to the nature of their spending, which is for highly broad and/or progressive spending, as opposed to the narrower and more regressive inflation tax outlays. When the government does pick specific winners, it's broad based and targets lower percentiles like with healthcare and other welfare programs, not specific white collar sectors of the economy.

1.3.5 Other Explicit Outlays

Tax: A public good, but a tiny portion of revenue. Credit Loss Provisions: Not an outflow into the broader economy, effectively just representing a reduction in revenue. Retained Earnings: Also not an outflow into the broader economy: it's what allows banks to relax regulatory constraints to continue inflation taxing. Dividends: Disproportionally flows into upper percentiles through equity ownership ([Federal Reserve Bank of St. Louis, 2025](#)).

1.4 Marginal Amplification

As previously noted, a 6.54% /yr tax and redistribution of an asset that's a minor portion of total net worth does not seem significant in a vacuum, even if it's regressive and imposed by a private entity. The real impact comes from spending habits. Lower percentile's asset mix is dominated by real estate and durable consumer goods, while upper percentile's assets are much more heavily weighted into equity ([Board of Governors of the Federal Reserve System, 2026a](#)), and upper percentiles spend a much higher portion of disposable income on investment over consumption ([Mian et al., 2025](#)). Since the inflation tax, broadly speaking, redistributes purchasing power from lower percentiles to upper percentiles, and those upper percentiles spend significant portions on an asset which lower percentiles do not own much of (even relative to individual portfolio net worth), and since prices are determined at the margins, small volumes of inflation tax profits can have out-sized impacts on upper percentiles net worth through equity prices, leading to significant inequality through asset inflation, especially when consistently compounded over decades. [Easterly and Fischer \(2001\)](#) present empirical evidence of inflation's impact on the poor showing "direct measures of improvements in well-being of the poor – the change in their share in national

income, the percent decline in poverty, and the percent change in the real minimum wage to be negatively correlated with inflation in pooled cross-country samples.”

1.5 Uniquely Unequal

Inequality is an unavoidable feature of reality, and capitalism accepts this, recognizing that unequal distributional outcomes are not inherently undesirable in of themselves, as long as resources are distributed in a maximally efficient and just way, such that everyone, or at least those that deserve it, are better off than they would be otherwise. But why is inflation tax induced inequality so uniquely corrosive, as opposed to many other structural forces/facts which may contribute to inequality but are still accepted in liberal democratic capitalist societies as unavoidable parameters of a system that makes everyone better off, relative to the alternatives? There are three fundamental reasons: 1) Inflation is a tax, in other words, an *involuntary* redistribution of material wealth (through purchasing power), which 2) is not collected by the government for the purpose of spending on public goods, and 3) is not collected explicitly in a way that makes citizens aware of what is happening, particularly who is getting taxed, who is taxing, and what they’re spending on.

For comparison, consider the simple capitalist force of wealth accruals from consumers to firms owned by wealthy investors who do not consume much themselves and are not wage earners of the firm. While a Marxist may argue this increases inequality and is evil, it strictly involves voluntary transfers of purchasing power. Wage earners are not forced to work at the firm, and consumers are not forced to spend on the firm’s products. This makes it incomparable to the inflation tax. Similar logic applies to the simple fact of compounding wealth: the fact that by just owning a large pool of money, one can earn compounding interest with no effort. Since wealth compounds, one who is able to save some spare money to invest is able to exponentially increase their wealth while those who do not have spare money to invest experience flat or linear wealth growth. Again this fact of reality does not involve an involuntary redistribution of wealth which results in inequality, making it incomparable to inflation tax. Another major source of inequality is simply the fact of being born with certain mental or physical gifts, or born into wealth. While being born into a certain situation is not voluntary, it was not the result of a forceful decision by any individual or institution, and is just a basic unavoidable fact of reality, again making it incomparable to the inflation tax (while conventional economists may similarly treat inflation as an unavoidable fact of reality, it is in fact a design choice). When con-

sidering involuntary redistributions of wealth which are accepted by society, inflation tax is *the only* process managed by a non-governmental/parental agent. This includes regular taxes, court orders, firm level decisions (such as layoffs, benefits, shareholder distributions, etc), and levies imposed by other private associations (HOA, unions, professional sports associations. The last two are voluntary because one enters into a contract before joining the association giving their consent to be governed by the association).

2 Core Distortion: Fixed Pies

Subsection 1.3 - subsection 1.4 demonstrate how inflation is a regressive tax collected by private depository institutions which redistribute profits disproportionately to upper percentiles, resulting in marginal liquidity into investment assets, particularly equity, disproportionately owned by upper percentiles, inflating their wealth at higher magnitudes than the lower percentiles's modest rate of asset depreciation (money holdings). But even though this artificially increases inequality, it's not immediately obvious how lower percentiles are directly harmed in a significant way, beyond modest depreciation of net worth.

A core problem with artificial-inflation-tax-induced-inequality is a result of fixed resource pies. While at a high level, the economy is a growing pie, it grows very slowly, and if the rate at which upper percentiles grow their slice by consuming lower percentiles slices outpaces the average rate of overall pie growth, then on average the utility of lower percentiles is decreasing over time. In this case, rising inequality is directly harmful, as if the pie were fixed. It's very difficult to determine, at the level of the entire economy, what portion of inequality is due to the inflation tax vs. what portion is due to other factors, how to correctly measure these concepts, etc. It's easier to identify effects in specific sectors of the heterogeneous economic pie, particularly those that have persistently slow growing supply and persistent demand: healthcare, childcare, education, real estate development.

"Baumol's Cost Disease" is one lens through which we can explain why sectors like healthcare, childcare, and education become so expensive with little productivity gains to justify price hikes (Baumol and Bowen, 1965). The reasoning is certain sectors, such as the aforementioned ones, are very labor intensive, fundamentally limited by how much capital investment, innovation, and automation can boost productivity: Despite technological improvement, hospitals are still extremely staff intensive.

Their work requires human skills that have proven difficult to replicate with technology. Even with the advent of artificial intelligence, it seems unlikely that chatbots will be able to replace nurses in an Intensive Care Unit any time soon, even though technology has been able to make specific procedures like laser eye surgery highly accessible. Childcare is more clearly restrained from automation; children need lots of devoted, personalized quality time and attention from an emotionally responsive human caretaker. Higher education is also very labor intensive, from admin and support to research and teaching staff. As a result, when wages for other more capital dependent/enabled sectors increase due to productivity boosts, wages for these sectors with slower productivity growth (typically labor intensive sectors) must also increase in order to stay competitive in the labor market. Since these increased labor costs did not come from productivity gains, they must be passed on to the customer through higher prices.

Baumol's hypothesis can be explained in general supply of goods/services terms rather than in labor market terms. For example, let's assume there is one sector like healthcare, or a collection of sectors like healthcare, childcare, education whose utility cannot be substituted with other sectors, and demand for which is consistent over time, and whose productivity gains are very slow. Let's assume spending for this category 50% for the average consumer. Say all other sectors have inherently higher probabilities of high productivity increases due more reliance on capital than labor, with simpler automation opportunities. In this situation, assuming the low productivity sector/set of sectors cannot be substituted with the high productivity sectors, with relative demand between the two categories remaining stable, then the spending breakdown will remain 50-50, but as the high productivity sectors increase goods quantities and quality, their cost per utility will decrease relative to the slow productivity sectors.

The supply version of Baumol's cost disease can also be applied to real estate, but it doesn't seem to be as labor dependent as healthcare, childcare, and education, making the classic labor market effect less applicable. Viewing it through the broader lens of supply however, it is clear that real estate is fundamentally limited in growth, like the other sectors. Economically useful developed land is in very short supply and cannot be increased easily. Even if you consider zoning restrictions as an artificial construct that has freedom to change, that is not the case for proximity to economically active zones, infrastructure access, etc. The factors that make a specific location valuable cannot simply be reproduced by inventing some innovative new way

to dig up land or erect a skyscraper.

The claim is not that these fundamental productivity growth rate limits between sectors can fully explain differences in relative inflation. There are many other complex and unique idiosyncratic factors which impact price. But the point is, accepting there are some critical sectors for utility maximization of the average consumer, such as the four aforementioned ones, with fundamental limits to productivity growth resulting in naturally high and rising relative prices, then the direct impact on lower percentiles from artificial-inflation-tax-induced-inequality becomes clearer. As the rich get richer through asset inflation powered by the stealthy and regressive inflation tax, administered by banks, their increased relative purchasing power allows them to bid for larger and larger slices of slow growing pies, exacerbating natural Baumol cost disease effects, and reducing the slices of lower percentiles. By contrast, wealth poured into high productivity areas may actually benefit lower percentiles as it can fund innovation and accessibility, such as for tech products like smart phones, where wealthy individuals provided the necessary early customer demand to enable eventual mass access. However if multiple critical sectors are deeply limited in their ability to increase productivity, and the extra growth which is enabled by the inflation-tax-funded upper percentile consumption does not outpace the rate at which upper percentiles out bid lower percentiles for larger slices, then this is another major indictment of the inflation tax.

Turning back to real estate specifically, we can see from [Board of Governors of the Federal Reserve System \(2026a\)](#) that while the trend is clearly a higher share of portfolio asset composition held in equity as wealth increases, upper percentiles still hold significant portions of their wealth in real estate. So maybe the inflation tax isn't so bad, because some of their inflation tax income goes into inflating real estate assets which helps lower percentiles, right? There are two main issues with this line of reasoning. First, single homeowners cannot effectively capitalize on their inflated house price. As [Green \(2025\)](#) points out: "Economists love the "Wealth Effect." They believe that when Zillow says your house you bought for \$200,000 is worth \$800,000, you feel rich and spend money. Let's look at that \$800,000 house. If you sell the house to "realize" your wealth, you are homeless. You must enter the market to buy a replacement [house]. But because all the [houses] repriced in correlation, your \$600,000 gain is immediately consumed by the purchase of the new, equally expensive house. The only way to unlock that wealth is to: 1. Die. 2. Downsize (move to a cheaper region, sacrificing income/opportunity/quality of life). 3. Borrow against it

(HELOC or reverse mortgage), which turns your equity back into debt. All three represent either a loss of utility or a conversion back into debt.” Second, the poorest of lower percentiles do not own homes at all, and real estate inflation only decreases their hopes of escaping rent serfdom.

3 Core Distortion: Overfinancialization

Aside from the inequality effects of the inflation tax, it results in instability and impaired productivity growth for the economy as a whole, by providing excess funding for unproductive activities, while profiting the financial sector engaged in unproductive and even actively detrimental intermediation. The results are persistent asset bubbles / inefficient distribution of limited resources, unless suppressed by significant counterforces (explained in detail for each time period). The modern era of consistent and significant monetary expansion begins with the creation of the federal reserve in 1913. I will briefly interpret this modern fiat era through the lens of the *inflation tax induced overfinancialization distortion*, as a surface level sanity check for the plausibility of the thesis. It is not meant to be comprehensive; a full scholarly in depth historical defense is outside the scope of this essay:

- *Creature from Jekyll Island (1913 - 1933)*: U.S. is on the gold standard, yet both outside and inside money supply expands: The Federal reserve reduces reserve notes to gold ratio from roughly 100% to roughly 40% to finance WW1. Various reserve requirements were greatly reduced in 1917 (Feinman, 1993, p. 587). From November 1914 - March 1920, commercial bank deposits grew at an average CAGR of 14.84% (National Bureau of Economic Research, 2026) (an incredibly rapid pace, average deposit ⁴ CAGR from Q4 1946 to Q4 2024 was 6.54%). Rajan and Ramcharan (2015) conclude that during the 1910s and 1920s U.S. farm crisis, agricultural credit availability “directly inflated land prices,” amplified the agricultural boom, and increased farmer indebtedness. Regional banks’ balance sheets become fragile. Monetary expansion also fuels the roaring 20s stock bubble (subsection 3.5), eventually triggering bank runs across the country, a crisis of course made worse than necessary by the ineptitude of the federal reserve (convincing banks to dismantle their previous decentralized

⁴Checking, savings, time and other deposits are aggregated by simple sum, since determining proper velocity weights was outside the scope of Kofman (2026). Ideally savings and time would have less weighting than checking.

clearinghouse liquidity support networks, and instead become reliant upon the federal reserve for liquidity support, while simultaneously encouraging addiction to the inflation tax drug, and then enforcing cold turkey at the worst possible time, to atone for “indulgent excess” by denying liquidity when the self-created crisis hits. The lesson learned is not that drug addiction is bad, but rather that drugs must be abused to get through a crisis induced by those very same drugs). The domestic gold standard ended in 1933.

- *New Deal (1933 - 1944)*: There is no room for bubbles to grow while the economy is in the initial stages of recovery. A slew of financial regulations are implemented: Federal Home Loan Bank Act, Glass-Steagall, Regulation Q, Securities Act, Securities Exchange Act, National Housing Act, Banking Act of 1935, Investment Company Act. The financial sector during the later stages is constrained from its natural bubble inflating tendency by these new regulations and heavy government intervention enforcing a war time economy. America has a uniquely massive industrial production mandate and ability as “the Arsenal of Democracy” for allied powers, which could absorb credit creation that would normally be excessive, flowing into asset bubbles.
- *Birth of the Eurodollar (1944 - 1971)*: The first Bretton Woods conference in 1944 established the dollar as the world’s reserve currency, pegging all currencies at a fixed rate, with the dollar still pegged internationally to gold at a fixed rate. International demand for dollars expands extremely rapidly as previously advanced economies rebuild, absorbing monetary expansion in place of asset bubbles, which are also suppressed by the new regulatory regime created during the new deal era and subsequently bolstered.
- *Fiat Standard (1971 - present)*: U.S. leaves the gold standard internationally, solidifying the federal reserve’s policy of monetary expansion accommodation by reserve expansion. The “Nifty Fifty” bubble pops, the first major postwar bubble formed at the end of the previous era. Stagflation is marked by poor economic conditions that suppresses the extreme level of monetary expansion from fully channeling into an asset bubble, resulting in blow off into unprecedented real goods inflation. Real estate begins its break from historical real price trends, driven primarily by land costs (the “investment asset” component of real estate, as opposed to the structure, which is a real good (Kofman, 2026, p. 26)), rather than structure inflation explained by construction costs (Knoll

et al., 2017, p. 331). Financial regulations got loosened and amended in the 70s: Regulation Q ceilings removed, Bank Holding Company Act Amendments, NOW accounts legalized, “May Day”, Securities Acts Amendments, *Marquette v. First of Omaha*. Financial deregulation accelerated in the 80s: DIDMCA, State-level thrift deregulation, Garn–St Germain Depository Institutions Act, SEC Rule 10b-18 stock-buyback safe harbor, SMMEA, Section 20 amendments by the Federal Reserve, etc. The commercial real estate bubble of the 80s forms, culminating in the S&L crisis. From 1990 - 1995, monetary expansion flatlines while the economy recovers from the S&L crisis, then the dotcom bubble inflates at the turn of the century. That bubble pops, the economy goes into recession, then housing inflation accelerates rapidly. That pops resulting in the GFC (Great Financial Crisis), followed by a very slow recovery. From then up until the present, there has been a combination of asset inflation and other inefficient distributions of resources, detailed in subsection 3.2 - subsection 3.4. Time will tell whether or not these asset inflations are persistent or bubbles. Candidates include real estate in general (*Bank for International Settlements*, 2026) (source only covers residential), The entire publicly traded stock market (*multpl.com*, 2026) (historically elevated Shiller’s P/E for S&P 500), driven by tech companies invested in the (potential) AI bubble, private equity and private credit.

Both the *Creature from Jekyll Island* and *Fiat Standard* eras are, under the inflation tax $\rightarrow$ overfinancialization / unproductive financial intermediation framework, are interpreted as the “natural” states of the economy under ever expanding credit, characterized by continual asset bubbles and/or other inefficient distributions of resources subsection 3.2 - subsection 3.4. Financial regulation (and broadly speaking other market distortions caused by government intervention) was/is either relatively nonexistent or reduced during these eras. The *New Deal* era is short and characterized by slow recovery and particularly strong government intervention, suppressing inflation tax’s natural bubble inflating tendency. The *Birth of the Eurodollar* era maintains and expands the government intervention introduced in the *New Deal* era, which again suppresses the natural tendency for bubble inflation, but is also further suppressed by the rapidly expanding international demand for dollars, as monetary expansion channels overseas rather than into domestic unproductive uses. While government regulation broadly and financial regulation specifically is of course necessary to some degree, and does suppress unproductive intermediation, it also suppresses

productive intermediation. This is not an argument for endless regulation to stop economic instability by suffocation, an unsustainable solution, rather the point is the natural outcome of an economy built on persistent monetary expansion is persistent asset bubbles and/or inefficient distribution of resources. Another important qualification is that asset inflation does not necessarily have to result in an unsustainable bubble. If the inflation tax redistributes from lower to upper percentiles, and those upper percentiles do not change their relatively high investment demand (vs real consumption demand) in a persistent way, then asset price inflation must be persistent. Those inflation tax flow profits channeling into investment have to go somewhere, and if one bubble pops then another must inflate, or in the case of land prices, they just continually rise higher and stay there. Unproductive intermediation comes at the cost of productive intermediation, which has underperformed since the early seventies (Gordon, 2014, p. 1). At the same time, the financial sector has done extraordinarily well. Christophers (2018, fig. 3) finds that in the early 80s, the financial + insurance industry's profit margin was around 10% and other industry's profit margin averaged around 3%, but the gap has widened to around 20% financial + insurance margins vs only around 4% profit margin for other industries, by the mid 2010s. Christophers (2018, fig. 2) demonstrates that while as a share of output, value-added, and full-time employment aggregates, the finance + insurance sector has drifted only from around 5% for all three aggregates in early 80s, to roughly 7.5% for value-added and output shares, and with no change in full-time employment share by the 2010s, the financial sector increased its share of corporate profits from roughly 13% to roughly 26%. Is this the state of a well functioning free market allocating resources with maximum efficiency? Or perhaps are there some major distortions rewarding unproductive financial intermediation? The following sections go further in depth about specific forms of unproductive financial intermediation that could plausibly be caused to some degree by the inflation tax.

3.1 Asset Bubbles

Why exactly does the inflation tax result in so much pressure for asset bubbles to inflate? The main mechanism referred to so far is the redistribution from lower percentiles, who must spend a large portion of their income on real consumption, to upper percentiles, who channel far more of their income into investment, and while the particular preferred asset may change over time (for example from treasury to tech stock), the demand for investment as a whole is fairly stable in the long run.

However, beyond the involuntary marginal liquidity extracted from bank account holders to bank revenue recipients, the inflation tax causes an additional, marginal, discretionary demand for investment, arising from the reduction in demand to keep money idle in deposits, as agents try to minimize their personal taxable base. Upper percentiles have lower liquidity needs and higher tolerance for volatility risk (they're not at risk of missing a rent payment if their investment goes bad), and can thus afford to reduce their taxable base more effectively. But doesn't increased marginal investment increase marginal growth? Yes, but it may be cancerous growth. Increased marginal investment is good if it's productive, if it's unproductive, there can be asset inflation, and potentially an asset bubble. One outlet is real estate, and increasing marginal investment into land isn't productive, it just increases rent seeking behavior, raising the cost of living and the cost of doing business. Since the *inflation tax initiated discretionary marginal investment* comes from investors who would have otherwise kept funds idle in bank deposits, on average, they will likely be more uninformed than the average investor in a world without inflation tax, since the inflation tax should not influence their interest in taking risk or spending time on researching profitable opportunities as much as it just forces them to preserve wealth, and thus makes them more likely to allocate limited resources to firms that won't use them productively.

3.1.1 Unsustainable National Debt

The marginal demand for investment caused simply by a need to protect wealth from inflation gets channeled to some degree into all assets, but disproportionately channels into the safest asset: U.S. Treasuries. If an investor is forced into investing simply to preserve their wealth from inflation, they will choose the closest alternative to a deposit, which is the safest and most liquid asset available. Other similar options include repo lending and Treasury - only Money Market Funds. The result is marginally cheaper debt financing rates for the U.S. government, arising not from the market naturally and freely deciding the U.S. government is a worthy borrower, but rather from decisions made under duress of the inflation tax. The price of U.S. debt becomes much less sensitive to the financial responsibility of the federal government, so naturally its spending becomes careless (in other words market discipline is deteriorated/distorted → national debt is mispriced), and eventually, careless debt fueled spending becomes an addiction the government *cannot operate without*.

One could argue market distortions / mispricings can be a price worth paying, that there's no such thing as normal or natural debt levels, and that it's more important

for the government to have enhanced ability to spend with impunity. However even under this framework, the fact remains that debt financing is fundamentally risky, and even the U.S. government isn't completely isolated from financial risk. The more U.S. government debt is issued without repayment, the more it becomes dependent on refinancing, and thus the more sensitive it becomes to interest rate policy of the Federal Reserve, which it can only influence indirectly. It also becomes more sensitive to investor confidence and the strength of the dollar's global reserve currency status. The MMT (Modern Monetary Theorist) claim that since the Federal Reserve can print dollars, which is the global reserve currency, it therefore never has to worry about debt, is not a serious argument. Hyperinflation has crippled many foreign governments that could print their own currency. If the only way a government can pay its own debt is by hyperinflating it away, this is practically equivalent to a default. Reserve currency status undoubtedly makes this less likely, but it does not make it impossible. In fact it's becoming concerningly more likely. All three major U.S. ratings agencies have downgraded the US government's sovereign credit rating from AAA. Central banks have been decreasing U.S. treasury holdings in favor of gold ever since around 2017, with the latter surpassing the former in 2025 for the first time since 1996 (Maharrey, 2025). It's foolish and irresponsible to blindly believe the dollar will remain the world reserve currency forever, no matter how much debt the government racks up or how much geopolitical leverage it loses. In a way, the world reserve currency is the Nash equilibrium of a coordination game (even though not all players equally benefit from the equilibrium, there are very high operational benefits to coordination), and while these equilibrium can be very stable for long periods, when they do change, it happens slowly, slowly, than all at once. Thus even slow deterioration of the dollar's reserve status should be taken seriously.

3.2 Zombie Lending

If banks were not able to satisfy borrower's residual liquidity thirst (originating from interest payments, as described in [subsection 1.2](#)) with new deposits as debt, then borrowers would have to satisfy residual liquidity needs with the excess liquidity of the recipients of bank revenue outflows (by becoming more productive and thus earning higher incomes, producing desirable consumer goods, attracting direct investment, etc.). Recipients would be presented with more opportunities to recirculate their excess liquidity back into the economy for use by productive borrowers. Instead, banks reduce the need for borrowers to pay back their debts through productive

investment, offering freshly minted money *ex nihilo*, decreasing the cost of borrowing above what is “natural”. Recipients do not recirculate as much of their liquidity back into productive use, and instead inflate asset bubbles and/or prop up unproductive “zombie firms”. [Banerjee and Hofmann \(2018, p. 70\)](#) find that zombie firms (“Firms with an interest coverage ratio less than one for three consecutive years and over 10 years old.”) grew from about 1% in the late 80s to just over 12% by 2016.

3.3 Ossification

Inflation-tax-powered centralization of capital into smaller and smaller groups of investing decision makers, slowly turns firm’s souls to bone by invariably expanding bureaucracy. Centralization of capital into fewer hands negates the core productivity engine of free markets: decentralized information. The marginal effect of centralized wealth from the inflation tax, is marginally richer upper percentiles whose marginal wealth is not owed to intelligent business/investment decisions, but rather by the privilege of being recipients of forced wealth redistribution by banks. This results in smaller and smaller concentration of investors who, year after year, are marginally more removed from the ground source of business realities, and whose effective aggregate brainpower decreases as wealth centralizes. Less brainpower is dedicated to innovating and perfecting founder-led small businesses, leading to declining relative market share ([Press, 2024, p. 2](#)) ⁵, more time is spent passive index investing ([Anadu et al., 2018](#)), with little thought spent on business fundamentals of the largest public companies.

Even more involved “smart money” from entities like venture capital, angel investors, private lenders, corporate/strategic investors, etc. become warped in their decision making as wealth centralizes. The more that capital centralizes into fewer and fewer hands, the more deals a shrinking pool of brainpower has to make, and thus must rely on generic and proven business doctrine, because they simply don’t have the time to deeply understand each market. If one person is in charge of 20 businesses, they physically do not have the time to understand the ins and outs of each firm’s unique circumstances, and thus the optimal decision to make. Each firm’s edge, their unique value proposition, are degraded. Yes this can be partially mitigated by hiring swaths of analysts and consultants etc, but ultimately they are still under the control of a few fund managers who call the big shots (to varying degrees depending on the type of investor of course, but the centralization directionality is

⁵The source cited as evidence of the empirical trend, not the cause.

the same). If that same number of analysts were each separately under the employment of 20 independent firms directly, building up domain expertise dedicated to each specific company for years, rather than rotating from one to the next every couple of months, strategists would develop decentralized expertise, providing fertile ground for innovation, rather than perfecting the same generic slide deck on how to cut costs by making the firm's products sh**ier and laying off employees.

The investor at the top does not have a true passion for any of the 20 companies, they aren't intrinsically motivated by building something truly great and innovative, which requires vision, dedication and the willingness to take a leap of faith. Creation, reshaping the industry, approaching the problem from a unique angle nobody's done before, those are secondary goals. While founders are no doubt also motivated by ROI, they're definitely more passionate about the creative process. For this reason, a founder is more likely to take a risk to improve product quality in hopes of expanding market share, while a manager of 20 businesses is more likely to take an alternate, safer, route of expansion: cut product quality to improve margins in order to increase capital for an acquisition of another firm. The first is a strategy of pie growth, the second of pie reduction and monopolization. This is of course is a crude generalization: every business should be cost efficient, and M&A's (Mergers and Acquisitions) can increase the pie through economies of scale / "synergies", but at some point, the anticompetitive and anti-innovation/agility/adaptability (a generalization) costs of higher dependence on generic cost cutting / M&A business strategies begin to outweigh the benefits. [Alfageme \(2025\)](#) argue that since 1980, in the U.S, M&A has increasingly functioned as a corporate growth strategy that substitutes for productive investment and helps drive larger firms, lower capex, greater market concentration, more concentrated profits, and higher markups.

Sprawling bureaucracy is another risk reduction method, imposing rigid top down control. The motive is maintenance, safety, predictability, structure, organization, etc. as opposed to free exploration and risk taking which lead to growth. As firms mirror their centralized investors by reducing in quantity and increasing in size, rigid top down bureaucracy expands. Firms feel there is no other way to effectively utilize their massive resources: they must be organized and controlled in a top down manner, otherwise, the fear is, there will be too much chaos. The founder is encouraged to step down, and a team of outside C-suite "managerial experts" come in to replace him. They begin introducing bureaucratic processes and middle managers, and are more likely to employ consultants and generic business optimization methods: cost/quality

cutting, leveraging, M&A, etc. While these measures are enticing because they're surefire ways to increase profits short term, if they're relied on too heavily in place of innovative expansion (doing more with the same existing resources, as opposed to just acquiring more existing resources from elsewhere), it will likely lead to long term brand degradation. The economic pie's growth will be stunted as long term sustainability, genuine expansion, and product quality are sacrificed for short term profits. And indeed Zook and Allen (2016, fig. 7) finds that from 1990-2014, founder led companies outperformed other S&P 500 companies by 3.1x. Of course there may be plenty of confounding variables, and founder-led companies do not necessarily mean less bureaucracy, but Zook and Allen (2016, p. 4)'s research concludes one of the key characteristics of the successful "founder mentality" is the "owner's mindset", characterized by "an antipathy to bureaucracy".

Of course every organization needs some degree of bureaucracy, and every business needs to be lean / cost efficient. The point is not that these characteristics are inherently destructive, they are not. The point is it's all about balance. These organizational behaviors/tendencies are more heavily relied on as capital centralizes, and capital centralization is out of balance, due to the inflation tax (not in totality but to a significant degree). Hamel and Zanini (2016, fig. 1) finds that from 1983 - 2013, "manager, business and financial operations" positions grew roughly 90%, while other occupations grew only roughly 38%.

3.4 Wealth Extraction

Private equity firms are the ultimate exemplars of what happens when inflation-tax-fed and thus overly centralized capital is able to aggressively leverage abundant monetary expansion to distort the market. A central pillar of neo/classical economic theory is the hypothesis that self-interested behavior, under conditions of *voluntary* exchange and competitive markets, channel private gain-seeking into the production of goods and services that others value. But the inflation tax is *involuntary*, as explained in subsection 1.5, and private equity firms exploit this fundamental flaw at the core of the economy to exsanguinate otherwise healthy businesses, leaving them as empty husks of their former selves. Generally speaking, the business model is centered around taking out massive loans at profitable rates to take over businesses, offloading the fund's debt onto the business, and aggressively improving margins by introducing economies of scale "synergies" through M&A, increasing bureaucracy (Obernberger et al., 2026) while laying off other roles, offshoring, and cheapening

products. The typical timeline before resale is short, typically around 5 years (at least it was. Recently, declining private equity demand is forcing funds to extend their ownership), so long term sustainability and health of the firm is not a priority. But how could this free market strategy be profitable if it leaves the economy worse off? While there is no unique involuntary exchange during the private equity process that one can point at as an obstruction to Adam Smith’s Invisible hand, it’s just yet another downstream consequence of the involuntary inflation tax, powering overly easy credit to be exploited through LBOs (leveraged buyouts).

Of course this is a major generalization; some private equity firms genuinely improve the business as opposed to exploiting it, but the fact that it’s uncertain which is the exception and which is the rule is concerning. There are many aspects to determining the improvement or degradation of a business, which include long term projections rather than just short term margins, available evidence does not paint a pretty picture: [Ayash and Rastad \(2019, p. 1\)](#) conclude “Tracking a sample of 484 public to private LBOs for 10 years after going private, we find a bankruptcy rate of approximately 20%, an order of magnitude greater than the 2% bankruptcy rate for the control sample”, [Bhardwaj et al. \(2025, p. 1\)](#) find “After accounting for positive selection, higher total debt (84% on average) dramatically increases the chance of financial distress (by 2.4 times the targeted firm mean) . . . They also reduce employee wages”. The healthcare industry is one of private equity’s most enticing targets, with strong, predictable, and stable cashflows, enabling them to withstand parasitic extraction. This is clearly particularly worrisome, as the evidence points towards exploitation of society’s most vulnerable: [Gupta et al. \(2024\)](#) find that in nursing homes, “PE ownership increases mortality by 11%”, and [Kannan et al. \(2023, p. 2365\)](#) find “After private equity acquisition, Medicare beneficiaries admitted to private equity hospitals experienced a 25.4% increase in hospital-acquired conditions compared with those treated at control hospitals”.

3.5 Big Bank’s World: Monetary Setup

Banking is one of the most, if not the most, heavily regulated industries in the U.S. Major portions of the modern regulatory framework (Glass-Steagall and Dodd-Frank) were constructed in the wake of two of the largest asset bubbles, one leading to the Great Depression, and another leading to the Great Financial Crisis. Other significant portions were also created in response to other bubble induced crises, such the S&L crisis. While conventional monetary expansion, in terms of bank deposits, was not

unusually high in the lead up to the Great Depression or GFC, internal ledger credits, specifically broker-dealer free credits (which is money outside of deposits (Kofman, 2026, p. 8)) expanded rapidly in the lead up to both, and contracted rapidly as the crisis hit.

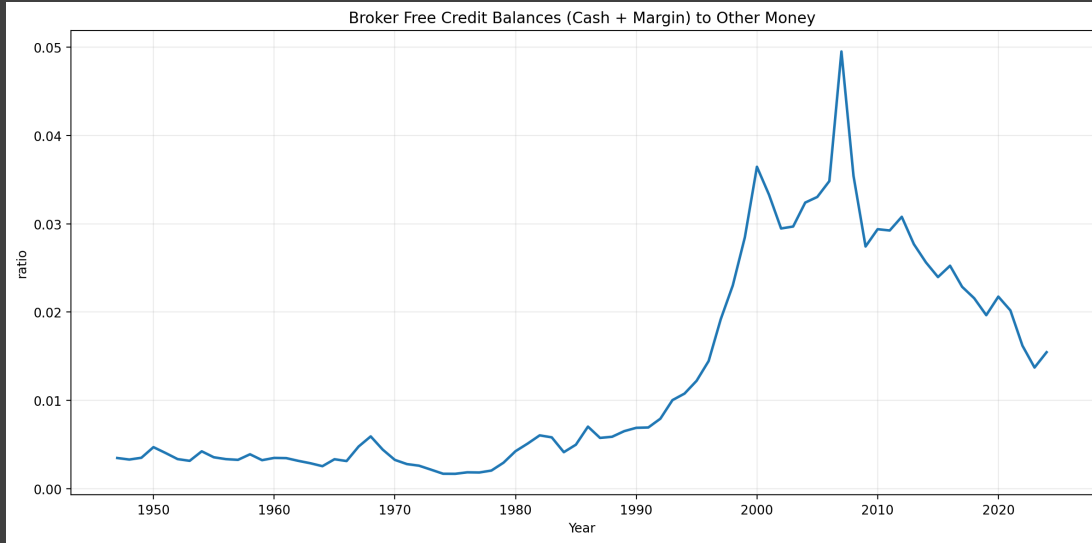


Figure 2. Ratio of outstanding *Broker-Dealer free credit balances (cash and margin)* over *incomplete, unweighted, simple sum monetary aggregate (checking + savings + time deposits + cash)*. Sources used for figure: (Board of Governors of the Federal Reserve System, 2025c,b,a, 1948–1970, 1981–2002; Financial Industry Regulatory Authority, 2026)

Senior Supervisors Group (2009, p. 10) note how at the onset of the GFC, mounting client concerns over creditworthiness “happened quickly at Bear Stearns, with client free credit balances declining drastically in the course of one week”. Consequently, “when a client of a U.S. broker-dealer withdrew balances from its account, known as free credit balances, the broker-dealer had to borrow to finance the remaining customer debits” (Senior Supervisors Group, 2009, p. 10). And because of the SEC customer reserve/lockup timing, “prime brokerage arms of firms subject to large customer withdrawals satisfied clients’ free credit balance withdrawals from the investment banks’ own liquidity until the next 15c3-3 lockup calculation was performed” (Senior Supervisors Group, 2009, p. 10). Months later, “The dynamics of the subsequent Lehman Brothers failure were similar to the Bear Stearns dynamics” (Senior Supervisors Group, 2009, p. 6), and “Following the failure of LBIE, [U.K] prime brokers received an enormous number of requests from hedge fund clients for the repayment of free cash balances and excess margin” (Senior Supervisors Group,

2009, p. 10).

Investigating the role of monetary expansion in the Great Depression is more difficult as the federal reserves did not have data on broker-dealer free credits prior to 1931. In general there is scarce financial data from this era, as many of the governmental data collecting agencies were yet to be established, or were not involved in extensive and comprehensive industry-wide data collection. However, as Woodlief Thomas testified to the Committee on Banking and Currency in 1934, “free credit balances . . . are pretty much the same as deposits. In other words, brokerage commission houses, in effect, do a banking business. They make loans and they accept deposits” (United States Senate Committee on Banking and Currency, 1934, p. 6451). Thus it would not be unreasonable to imagine a rapid monetary expansion and subsequent “bank run” as a major contributor to the Great Depression, just as it was for the broker-dealer arms of Bear Stearns and Lehman, especially considering the looser/nonexistent regulations on commingling customer funds. Indeed, Eiteman (1933) argues that a major source of broker margin account funding was not only broker loans from banks/corporations, but also from other customers: holders of free credit balances. He argued that published brokers’ loan statistics were misleading because they excluded customer credit balances, for which the differences from outside bank/corporation funding were economically minor, from the broker’s point of view. In March 1929, Eiteman (1933, pp. 458–460) used NYSE transaction-volume data and the monthly increase in brokers’ loans to estimate that net cash withdrawals offset only about 1% of the credits generated by customer stock sales. But if outstanding broker-dealer free credits were so large, then why did broker-dealers not fail *en masse* upon the onset of a free-credit “bank run”? In fact, only 0.16% of stock exchange members failed in 1929, even below the 0.27% average from 1922-1928 (Rappoport and White, 1991, p. 21). There are two main possible explanations: 1) margin customers liquidate, selling to free credit holders. Free credits and margin debits cancel each other out, contracting the monetary supply of free credits without a bank run. 2) Those withdrawals that did occur, which were only 7% of sales-originated customer credits from January 1929 - April 1930 (Eiteman, 1933, p. 459), could also be met by transactions where brokers aggressively and quickly liquidate margin accounts, or by voluntary margin liquidations, in which the end buyer paid with deposits/cash. The broker would use those cash proceeds to payout free-credit withdrawals. Thus it’s entirely plausible that the roaring twenties bubble was fueled by rapid monetary expansion, whose popping was triggered by, or subsequently caused, a rapid mon-

etary contraction, just as in the housing bubble in 2008, only without significant broker-dealer failures, the major sources of extraordinary monetary expansion.

Thus excessive, unstable, and unproductive monetary expansion likely fueled both the roaring 20s bubble and the subprime bubble. Ultimately, as a result (in combination with other factors), hundreds of banks failed during the GFC, while thousands failed during the Great Depression. In both cases, failures were primarily concentrated in smaller banks (McCord and Prescott, 2014), (Wheelock, 1995, pp. 30–31), while larger institutions consolidated their relative market power, with the infamous bailout of systemically important institutions during the GFC. The potentially longer term impact however, was the regulatory response. Rather than directly addressing what I have attempted to argue is a core issue, inevitably unproductive monetary expansion, mountains of banking regulations were implemented, which are constrictive for existing banks and make starting and maintaining a new bank an extremely long, costly, and difficult process. Big centralized banks have risen to dominate, while small banks struggle (from 1984 - 2011, the share of U.S. banking assets held by community banks declined from 38% to 14%. The number of federally insured banks decreased from 17,901 to 7,357 (Federal Deposit Insurance Corporation, 2012, p. 1)). Big banks become so big we cannot afford to let them fail. This is not to say that banking regulation is unnecessary. It is required if we must operate under a system where banks have the incredible discretionary power to expand inside money supply endlessly, accepted and encouraged by the federal reserve's official policy of accommodation (2% inflation target) by outside money expansion. Even with a fixed monetary base there would need to be some basic level of regulation. However the mountain of banking regulation the economy currently operates under is only unavoidable to the extent that we refuse to consider a more resilient monetary system.

3.6 Big Bank's World: Consequences

But could it actually be beneficial to have a highly centralized banking system, as big banks can afford to be more diversified? Small/medium size banks generally must specialize in specific areas to remain competitive against economies of scale (for example concentration in specific regions), making it harder for each individual bank to become diversified, which could degrade their core competitive advantage. If you take the collective balance sheets of many small regional banks and instead combine it into a single balance of a large bank, the odds of a bank failure have decreased due to diversification, without a change in the risk profile of any single

loan. By reducing the chances of a bank failure, contagion risk has been reduced, which was a major crisis amplification channel during both the great depression and GFC. The banking system is now more resilient, correct? Not necessarily. Now you’ve created a system with banks that are Too Big to Fail (TBTF), and they are well aware of that. As Gary H. Stern and Ron J. Feldman point out in their 2004 book “Too Big to Fail: The Hazards of Bank Bailouts”, the TBTF precedent was set well before 2008, during the distresses of: First Pennsylvania Bank, Continental Illinois, MCorp, First Republic Bank, Bank of New England, First National Bank and Trust, and many S&L thrift failures (Stern and Feldman, 2004, pp. viii, 13, 30–31, 152–153). In the largest of these cases, uninsured depositors/creditors were bailed out due to concerns over systemic risk (Stern and Feldman, 2004, p. 152). Perhaps the most blatant signal was the Comptroller of the Currency’s congressional testimony, which was summarized by the Wall Street Journal’s report: “U.S. won’t let 11 biggest banks in nation fail . . . Testimony by comptroller at House hearing is first policy acknowledgment” (Stern and Feldman, 2004, p. 31). The fundamental resulting issue, the authors highlight, is TBTF creditors do not impose adequate risk discipline because they expect to be bailed out (Stern and Feldman, 2004, pp. 11, 17–19, 24–25). Even though equity holders still stand to lose, they’re making a completely different risk calculation. Equity holders reward risk if it comes with high returns, and are more diversified (Stern and Feldman, 2004, pp. 24–25, 30–32). Risk thus becomes mispriced, which could help to explain the extreme levels of leverage and counterparty risk large institutions took on during the GFC. The authors argue, and were soon proved correct, that FDICIA (Federal Deposit Insurance Corporation Improvement Act) did not solve TBTF because it merely formalized the bailout process which was already implemented in prior cases (Stern and Feldman, 2004, pp. 78–79, 149–150, 154–157). Thus bank centralization transforms *small bank failure* → *contagion risk* into *TBTF systemic risk*.

The implementation of new and highly robust Basel III capital and liquidity requirements under Dodd-Frank, including enhanced regulation targeted at GSIBs (Global Systemically Important Banks), may seem to have, on the surface, successfully addressed the TBTF issue. But when delving deeper, when reaching into the shadows, it seems increasingly likely that detrimental effects caused by fundamental structural realities of big bank dominated systems cannot be adequately addressed without decentralization. Basel regulation does not come without significant market distortions, nor is it clear that risk is substantially reduced, though it is transformed.

3.6.1 NDFI Feast

As banks, particularly GSIBS, are increasingly more restrained from risky lending (as dictated by generalized and rigid regulatory risk weights) post GFC, they turned to NDFIs (Non Depository Financial Institutions) to fill their risky business lending gap.

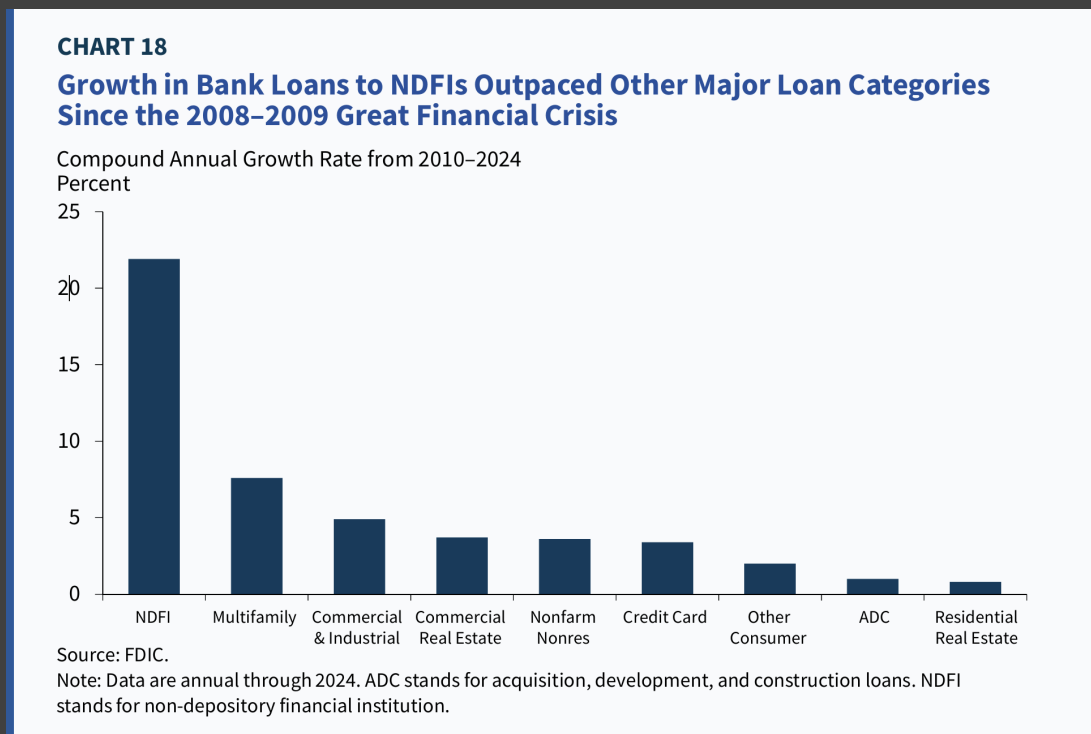


Figure 3. Extraordinary post-GFC growth in bank to NDFI loans. Source: (Federal Deposit Insurance Corporation, 2025, fig. 18)

NDFI is a diverse category, including private equity funds, mortgage lenders, private credit funds, insurers, Real Estate Investment Trusts, Business Development Companies, securitization vehicles, etc. These agents are sometimes referred to as “shadow banks” as they perform bank-like financial intermediation functions by pooling funds and redistributing them into investment opportunities in the real economy, including to Small and Medium size Enterprises (SMEs). Their profit generating utility can be described as maturity/risk/liquidity transformation, which are the core functions of a traditional bank. Generally speaking, banks prefer lending to these middlemen rather than engaging in direct lending partially because regulatory risk weights require less capital to be held against NDFI loans (as opposed to direct Commercial and Industrial (C&I) loans for example), as they provide an additional risk

buffer due to their expanded funding capacity and balance sheets outside of the bank. As [Federal Deposit Insurance Corporation \(2025, p. 24\)](#) notes, “factors underlying the growth in NDFI loans include banks seeking to reduce risk in direct lending in certain types of loans post-GFC”, because “The composition and structure of bank loans to NDFIs generally exhibit a lower degree of credit risk. Most of the credit facilities to NDFIs are short-term revolving credit lines that are typically collateralized with conservative advance rates against the collateral pledged, providing a layer of loss protection. Supervisory observations reflect strong historical performance and more favorable credit ratings for banks’ loans to NDFIs compared to traditional commercial loans” ([Federal Deposit Insurance Corporation, 2025, p. 24](#)). Unsurprisingly, the majority of this growth has been led by more heavily regulation-constrained GSIBs. In the second quarter of 2019, the four largest banks (JPMorgan Chase N.A., Bank of America N.A., Citibank N.A., and Wells Fargo N.A.) held about half of NDFI loans outstanding ([Fritzdixon, 2019, p. 38](#)). By December 31, 2024, 87% of the NDFI balance was held by banks with assets over \$100 billion ([Federal Deposit Insurance Corporation, 2025, p. 24](#)). As mentioned earlier, NDFIs are a diverse category, yet “Most of the growth in bank loans to NDFIs has been led by loans to private equity and private credit entities” ([Federal Deposit Insurance Corporation, 2025, p. 24](#)). Furthermore, private credit fund’s assets are heavily concentrated in loans to private equity funds ([Becker Friedman Institute for Economics, 2026, fig. 1](#)). The end result is that post-GFC regulation, particularly the heavy GSIB regulation, has resulted in a market distortion which provides an unnaturally large funding capacity for the wealth extraction behavior outlined in [subsection 3.4](#), further degrading the “invisible hand” defense of exploitation (when it occurs, again not all private equity firms aggressively abuse their regulatory arbitrage advantage).

3.6.2 Encroaching Shadows

The post-GFC accelerating shift of bank (particularly GSIB) intermediated C&I loans into shadow banking should affect SMEs more than larger firms because they’re generally riskier. [Favara et al. \(2021, p. 1427\)](#) finds that banks subject to higher GSIB surcharges reduced the dollar amount of loan commitments to nonfinancial firms relative to other banks, including small and speculative grade firms. However this does not seem to influence firm’s borrowing capacity, as their borrowing demand was replaced by non GSIBs, “without experiencing much impact to their leverage and real investment” ([Favara et al., 2021, p. 1427](#)). Small business debt markets experienced

significant change: large commercial bank’s small business lending volume fell by 45% since the GFC, and the number of loans hadn’t recovered to pre-crisis levels by 2021, with the decline being the most significant for the top 4 banks (Bank of America, Citigroup, JPMorgan Chase, and Wells Fargo) (Erel and Inozemtsev, 2025, pp. 1237–1238). However this supply gap, along with the broader decrease of all bank lending to small business, was “almost perfectly offset” by the increase in finance companies and FinTech lending (Gopal and Schnabl, 2020, p. abstract). Davydiuk et al. (2020) find the dramatic rise of nonbank corporate middle market direct lenders (private credit), in particular Business Development Companies, can be attributed to the shortage in credit supply left by traditional lenders in the wake of post GFC regulation. But the funding adaptation for SMEs may not have been as simple as *less direct bank loans* → *more NDFI loans*. It could look more like: SMEs lose access to some bank funding, but do not immediately get offers from NDFI lenders. However business models are still fundamentally sound, so they do get buyout offers from private equity funds, and only then do they get access to adequate NDFI funding thanks to the industry connections of private equity funds with private credit funds.

Setting that hypothetical aside, why does it matter that SMEs become more reliant on shadow banks? After all, shadow bank funding can have advantages such as more flexible terms, more industry specific expertise, convenience, etc. While this may be true, and indeed a healthy market could include both bank and shadow bank options, shadow banks also come with macroprudential financial stability tradeoffs. The intent of banking regulation is to reduce this risk, but a significant portion of that risk may have just migrated into the opaque and unregulated shadows thanks to regulatory arbitrage, a possibility that financial regulators are aware of and increasingly concerned by (Berrospide et al., 2025), due to the increased risks of looser underwriting standards (Martin-Buck, 2019, p. 41).

Beyond concerns involving just direct lenders, NDFIs broadly are more reliant than banks on unstable funding sources without explicit government backstops (such as deposit insurance for banks), making them more vulnerable to runs (for example mutual funds when investors attempt to redeem shares) (Erel and Inozemtsev, 2025, pp. 1253–1256). The increasing interconnectedness of NDFIs and banks may make the banking system more resilient to a failure of any single bank, and could decrease the risk from uninsured deposit runs, but could also increase the odds of systemic instability in the case of a large enough shock originating from the shadow banking sector (Erel and Inozemtsev, 2025, pp. 1256–1257). The layer of separation

between banks and the end borrower provides additional balance sheet and funding cushioning, as mentioned earlier, but it also “makes the evaluation of counterparty risk more challenging and reduces the financial stability of the financial system, as market participants are unlikely to internalize the impact of this change” (Erel and Inozemtsev, 2025, pp. 1256–1257).

3.6.3 Big vs. Small

In subsection 3.5 I argued that our contemporary centralized banking system was birthed from major financial crises, which in turn find their roots in unproductive monetary expansion. Subsection 3.6.1 - subsubsection 3.6.2 demonstrate why it’s not clear that our highly centralized banking system can truly benefit from the resiliency of diversified balance sheets without imposing a burdensome regulatory framework which potentially re-creates significant risk in the shadowbanking sector and distorts capital markets in favor of private equity funds (as opposed to more decentralized direct C&I borrowers). In order to fully evaluate the ramifications of monetary expansion from the viewpoint of banking architecture, aside from financial stability, the other fundamental factor to consider is productivity growth.

For the same fundamental reason centrally planned economies underperform free market economies, centralized banking architecture should be expected to underperform decentralized banking architecture, from the perspective of productivity. To the extent that firms are dependent on external financing, Bankers (and the most bank-like shadowbankers) are similar to central planners, operating at the highest level of economic abstraction, as they decide which projects deserve development. They must take a bird’s eye view abstracted away from the low level details of any single firm’s day to day operations, formulating a broad investment strategy including many projects. The degree to which they’re centralized and constrained by stiff bureaucratic decrees/regulation, is the degree to which they resemble the USSR’s Gosplan officials. For example, one big bank of 300,000 employees managing 3 trillion in assets does not have the capacity to process as much decentralized local market information as 1,000 small banks, each with 300 employees managing 3 billion in assets. The hierarchical nature of firms results in a significant burden placed on one small team of c-suite executives to make the highest level decisions for the single large bank, compared to 1,000 senior management teams of the small banks.

Since large banks have less information processing power at higher decision making levels, when making C&I decisions, it’s more efficient from a human capital perspec-

tive to execute fewer numbers of larger deals with larger firms. This encourages M&A/LBO activity, being some of the largest deals a bank can make, whether to hold the assets on their own books or, increasingly, to offload risk to shadow bank entities (Erel and Inozemtsev, 2025, p. 1242) while collecting fees based on deal size. One option is the leveraged loan, a term used to describe deals made with lower-rated nonfinancial corporate borrowers, which typically have high debt levels and use it to finance M&A/LBOs (Martin-Buck, 2019, p. 44). In the first half of 2019, the largest banks arranged more than 92% of leveraged loans (Martin-Buck, 2019, p. 46). These loans have become increasingly more risky, as the share of “covenant-lite” loans have decreased from around 0% in 2000 to 85% of new loans in 2018. Reported leverage for borrowers has also increased from 3.9 earnings in 2010 to 5.4 times in the first half of 2019. “Actual leverage could be even higher, as the use of earnings “add-backs,” which inflate earnings to account for future anticipated cost savings or revenue increases, has become more prevalent” (Martin-Buck, 2019, p. 44). Furthermore, “Bank exposure to risk from nonbanks that participate in leveraged lending is opaque, and the nature and size of the risk is obscured. Risk is difficult to quantify because it is not reported in a standardized manner” (Fritzdixon, 2019, p. 36). Of course risk isn’t inherently negative, but is it justified by reward? Maybe not, as “Research shows roughly two-thirds of M&A transactions destroy shareholder value for the acquiring companies” (Mauboussin, 2010, p. 3). We should question if the current levels of M&A/LBO activity are really the most efficient distribution of resources, or if they’re the byproduct of a big bank’s world shaped by inflation tax and resulting regulatory market distortions, acting against the invisible hand of free markets.

The importance of small nonfinancial firms is a critical piece of the story as well. Audretsch and Thurik (2001) find that higher entrepreneurship, measured by the relative share of economic activity accounted for by small firms and self employment rate, consistently tends to result in higher growth rates and a reduction in unemployment among multiple OECD countries over different time periods. Unsurprisingly, young firms have higher employment growth, sales growth, and productivity rates than older firms when controlling for firm size, country, industry, and year fixed effects (Ayyagari et al., 2011, p. 4). However when adjusting for age, while smaller firms do have higher employment and sales growth than larger firms, they have significantly lower productivity growth (Ayyagari et al., 2011, p. 14). This does not invalidate the findings of Audretsch and Thurik (2001), but rather suggests that while on average, smaller firms (when adjusting for age) are less productive, there’s a subset of innovative young and

initially small firms which are critical for productivity growth in the macroeconomy, and eventually grow large, maintaining and increasing productivity as they expand. This is why the rate of dynamism (the rate of economic change in terms of firm creation, growth, and destruction) is so critical. A highly dynamic economy is one in which creative space is ample, where ground is fertile for risky experimentation and thus innovation. A stale, stiff, high-startup-cost, entrenched economy, which may even be dominated by a few large but highly productive firms, is not fertile ground for growth, even if productivity may be high at one given point in time. Neither is an economy clogged up with a mass of unproductive small firms. The importance of small firms for growth and their relative average unproductiveness is not a contradiction, and explains why “high income countries are characterized by high rates of entry and turnover of small firms rather than a large SME sector” (Ayyagari et al., 2011, p. 3).

The issue from a lender’s perspective is the uneven distribution of success among small firms, particularly young ones. Many will file for bankruptcy, while a few will be hypersuccessful. Yet traditional bank originated debt contracts don’t utilize warrants, and thus their upside is a predetermined single or low double digit interest rate, while their potential downside is 100%, making traditional loan structures unattractive for such high risk lending (More flexible nonbank lenders provide value by filling this gap (Erel and Inozemtsev, 2025, p. 1245)). The asymmetric advantage required to overcome this incentive misalignment is “soft information” (information that cannot be credibly communicated from one agent in a firm to another). Small banks are structurally better suited to take advantage of “soft information” than large banks, giving them a competitive advantage in small business lending, which explains the associated empirical trend wherein large banks are less willing to lend to informationally “difficult” credits (Berger et al., 2002, pp. 20–21), (Mkhaiber and Werner, 2021). Thus a decentralized banking system is more likely to support dynamism, by lending to innovative entrepreneurs / small firms, which is critical to macroeconomic productivity growth. However banking is still an inherently risky business with externalities affecting the entire economy, so some level of critical/efficient regulation (particularly balance sheet transparency) is necessary. This is one reason why our current solution of encouraging large centralized banks to support large firms while leaving SMEs to depend on unregulated and shadowbanks is inadequately unstable.

Duan et al. (2023) argue China’s meteoric rise is an empirical example of decentralized banking’s power, fueled by the architecture designed under Deng Xiaoping.

Mainstream economics has struggled to explain the success (Duan et al., 2023, pp. 2–4), dubbing it one of the “Asian miracles”. Under Mao, China followed a highly centralized banking model based on the Gosbank, stifling growth (Duan et al., 2023, p. 17). Under Deng Xiaoping, this model was rejected, and thousands of city and rural banks were created (Duan et al., 2023, p. 10) (Prior to 2000, there were over 50,000 rural credit cooperatives (Asian Development Bank, 2016)) which Duan et al. (2023, p. 16) argues was one of the two fundamental keys to success, since these banks “act as the engine to credit-fuel the liberated industrial complex: local entrepreneurs in SMEs and the enlivened SOEs”. Deng Xiaoping himself contended his “banking and credit guidance reforms were the ‘secret of success’ and the main explanatory variable for the unprecedented economic expansion in China” (Duan et al., 2023, p. 17). On the flipside, Duan et al. (2023, p. 18) contends the deliberate killing of over 5,000 banks in the last two decades is a major factor in Europe’s economic decline.

4 Core Distortion: Basel’s Other Side Effects

As argued in section 3/subsection 3.5, monetary expansion inevitably leads to unproductive investment, which occasionally leads to major financial crises, which in turn leads to suffocating banking regulation, focused on safety over growth. Current frameworks behave like palliative treatments, targeting symptoms (financial crises) rather than the underlying tumor (inflation tax), and bring corrosive growth-stifling side effects affecting the intermediation of all banks broadly, beyond the compositional effects of big bank’s competitive advantages.

A particularly strong example of the downstream ‘safety over growth’ regulatory tradeoff decision was implemented in the U.S. with financial regulator’s introduction of Basel I risk-weighted capital requirements in 1989, which weighted one-to-four family residential property mortgages at 50%, and most “claims on private obligors” at 100% (Federal Deposit Insurance Corporation, 1989, p. 11517). Since C&I loans became twice as capital intensive as residential mortgages, this artificially changed the risk-reward calculation of banks, encouraging them to inflate real estate prices rather than enabling productive economic growth (credit creation for unproductive asset purchases leads to asset price inflation, while credit creation for productive business investment leads to economic growth (Werner, 1997)). Empirically, it seems that banks indeed modified their strategies in response to the new rules, breaking from historical trends just around the time of Basel I’s introduction (of course correlation

does not mean causation, but the timing is striking). Prior to 1989, Mortgage and Nonmortgage lending trends tracked each other closely. From 1989 - 2011, Mortgage lending skyrocketed while Nonmortgage lending stagnated (Jordà et al., 2014, fig. 2). After 2011, this trend seems to have continued, and potentially even intensified (Hanson et al., 2024, figs. 2, 3, 7).

Another other major regulatory safety over growth tradeoff comes with Liquidity Coverage Ratios, introduced post GFC. These requirements favor liquid securitized loans over illiquid direct loans (in case of a sudden panic driven liquidation event), resulting in bank disintermediation for “informationally sensitive” loans ⁶, (loans which cannot easily be bundled into securities, unlike mortgages, auto loans, credit cards, and large corporate debt, and of which the borrowers typically cannot issue their own securities). Buchak et al. (2024) find that the share of informationally sensitive loans (defined as “bank balance sheet lending”) has secularly declined relative to all credit market loans (including securities). Non-mortgage informationally sensitive loans were around 38% in the mid sixties to late 80s, declining to around 30% by 2022 (Buchak et al., 2024, fig. 3). Buchak et al. (2024, p. 6) partially attribute this trend to regulatory pressures on banks, including but not limited to liquidity coverage ratios, “prompting them to pivot towards securities rather than extending credit” , supported by temporal correlations between stricter regulatory requirements and declines in informationally sensitive lending (Buchak et al., 2024, p. 4).

Naturally all regulation has come with tradeoffs, but effective regulation should have minimal side effects relative to the problem it solves, otherwise it doesn’t pass cost benefit analysis. I believe there is enough evidence to suggest that financial stability in the U.S. has not been solved with the current regulatory framework (with the rise of far less regulated and opaque shadowbanks), while productivity has secularly stagnated (2.36% a year for eight decades prior to 1972, 1.59% a year from 1972 - 2014 (Gordon, 2014, p. 1)), as well as dynamism (“Since the early 1980s, various measures of entrepreneurial activity in the United States have seen a secular decline” (Decker et al., 2016, p. 1)). Both of these trends have partially reversed post pandemic, but have not yet fully reversed the longer run decline (Congressional Research Service, 2025), (Decker and Haltiwanger, 2024). Of course there are countless potential variables to consider, but there is reason to believe the U.S. implementation of Basel is a significant one. If the two aforementioned side effects are considerably costly,

⁶This term refers to contracts dependent on “soft information” (Buchak et al., 2024, p. 21), which include smaller and younger corporate idiosyncratic corporate borrowers (as mentioned in subsubsection 3.6.3).

then that is yet another reason to believe the downstream consequences of monetary expansion are not insignificant, and have not been adequately neutralized.

5 Core Distortion: Data Degradation

A price for a given good G is a highly informationally dense number, as the summary of all market transactions for G . In theory, that price can be reasonably interpreted as the relationally objective value (objectively correct relation to other prices) of G , based on the combination of relationally subjective valuations of marginal buyers and sellers. However, for a price to truly be the most accurate, relationally objective value of G that is possible, the market decisions of marginal buyers and sellers must be as well informed as possible, so that they maximize *actual* utility as much as possible rather than *expected* utility. In other words, their utility maximization must be as least constrained by limited information as possible. Since prices are relative, meaning they exist in a system of relations to other prices, important components of the information required for a marginal buyer's decision to maximize their actual utility from the consumption of a particular good, are the prices of other goods. Prices are data, but data requires interpretation before it can be acted upon. Now take the following scenario: the entire economy is in equilibrium at t_0 . All prices are completely relationally objective measures of value. At t_1 the supply of money expands when an agent increases their endowment with newly created money, and spends it distributed across goods H , J , and K , according to their internal preferences / utility function which have not changed. This puts upwards pressure on the prices of H etc., since nominal demand has increased. At t_2 The marginal buyers and sellers of G are informed of the increase in the price of H etc. at t_1 , but they incorrectly interpret the increase in price of H etc. as some combination of: a change in the internal preferences of marginal buyers and sellers of H etc., or an increase in the money endowment of a marginal buyer of H etc. *acquired due to a transfer of money endowment from another market participant*. Essentially, they do not account for the increase in money supply as partially responsible for the increase in price of G (along with the fact of how the new money was distributed specifically to a marginal buyer of H etc.). If they were correct, then solely from the provided information there is no direct reason for the price of G to adjust, since the combination of the two considered possibilities truly represents an increase in relative value of H etc.. But since the change was impacted by the increase in money supply, then to the extent that the price

of G is impacted by future expectations, the marginal buyers and sellers of G should have adjusted their internal preferences / valuations / utility functions involving G to account for the increase in money supply (which will eventually circulate its way into the marginal buyers and sellers of G). To the extent that the price of G is not impacted by future expectations, then while it may not be utility maximizing for the price of G to adjust immediately (for the marginal buyers and sellers of G), the relative prices of G to H etc. will be divorced from their relationally objective values (which can cause deviations from utility maximization in other markets which are informed by the prices of G and H etc.), until eventually the price of G properly adjusts to reflect relationally objective values as new money circulates. In either case, G becomes temporarily mispriced due to the expansion of money, since some information about market good's relationally objective value is lost when compressed into prices, as changes in money supply are combined with other changes in demand related to idiosyncratic facts specific to each goods utility relative to other goods, along with changes in each good's idiosyncratic quality adjusted supply. The resulting mispricing results in suboptimal distributions of resources. Preventing it depends on the ability of market participants to: when presented with a change in price, correctly separate two fundamentally different causes: changes in factors related to idiosyncratic facts about relative utility and quality adjusted supply, vs. changes in money supply.

Changes in idiosyncratic facts are more intuitive; they are the natural, default, implicit internal interpretation of a change in price. If the price of a car, a house, or a stock increases, the immediate conclusion might be "other people want more of this good and are willing to pay higher prices to get it, so that means its objective value based on how much utility it provides has increased", a second thought might be "the intrinsic utility of this good hasn't changed, but it's become more rare and thus more valuable". Both of these instincts' ultimate conclusion is the specific good's relationally objective value has increased. But this is not necessarily the case, due to a third possibility: the supply of money has increased, and flowed into that specific good before others. Unsophisticated market participants should be expected to be more likely to fail to account for the third possibility, and thus more likely to misprice goods and thus stray from their utility maximum. Thus inflation tax is also a regressive financial illiteracy tax.

Sophisticated market participants aren't immune to mispricing as a result of monetary expansion either, since it's very difficult to accurately estimate. There are many subjectively-influenced modeling judgments to make. Some conventional frameworks

estimate the effects of monetary expansion on prices in their inflation models without even directly including monetary expansion. There’s uncertainty about the theoretical underpinnings of models themselves, yet alone how to properly determine parameters of those models. Additionally, beyond measurement errors of monetary expansion *ex post*, accurate pricing depends on even more unreliable *ex ante* predictions, particularly for financial assets and other goods dependent on future expectations.

Essentially, monetary expansion introduces an additional variable affecting the price of goods (as opposed to a fixed money supply), which makes market distortions / suboptimal allocations of resources more likely because of informational processing limits of market participants. If money supply was fixed, secular general inflation would have a simple constant product relationship with the quality-adjusted supply change rate of all goods (Kofman, 2026, p. 21), making one directly interpretable from the other.

6 Core Distortion: Financially Illiterate Wage Bargaining

It’s no secret that the average voter despises unexpectedly high inflation. However Shiller (1996, pp. 2–3) finds that even low/steady inflation is generally disliked, and found that one of the primary reasons is belief in a “sticky-wage model” whereby wages systematically lag corporate profits, leading to an ever decreasing labor share. While Bach (1974) and Alchian and Kessel (1960) have been cited to refute the persistent redistribution towards business profits hypothesis, they do not settle a more nuanced bounded-lagging-stair vs forward-looking-stair issue. Given an average, equilibrium labor market wage level in a frictionless price world, that would rise continuously and steadily to track general inflation, the issue at hand is: do actual nominal wages stay flat as inflation rises and jump up suddenly at a renegotiation/job-switch to the “fair” wage level (bounded-lagging-stair), or do wages jump above the current equilibrium wage level at renegotiation, to compensate for the real wages lost to inflation from the previous renegotiation interval (forward-looking-stair).

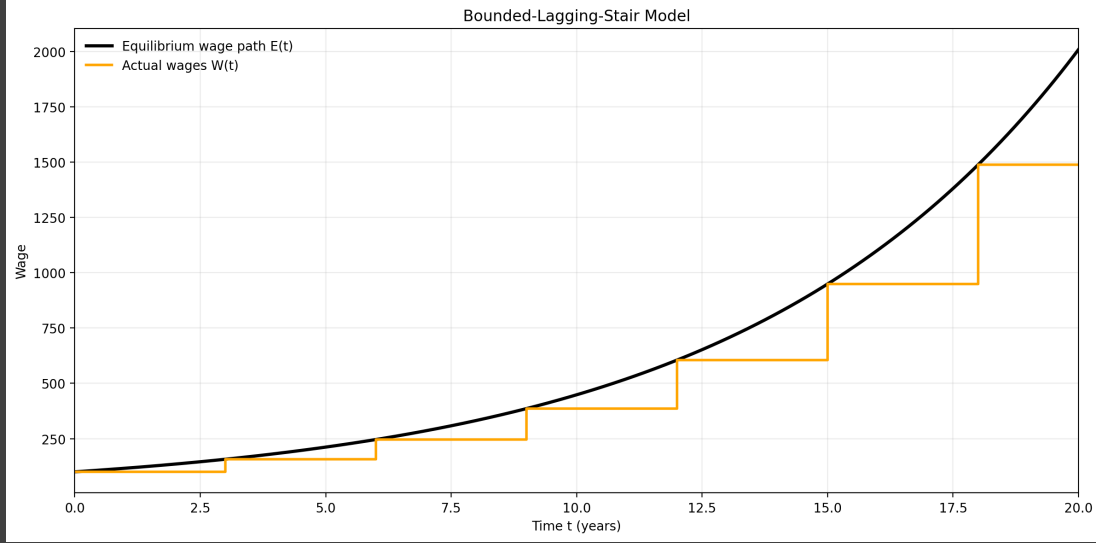


Figure 4. Bounded Lagging-Stair Model.

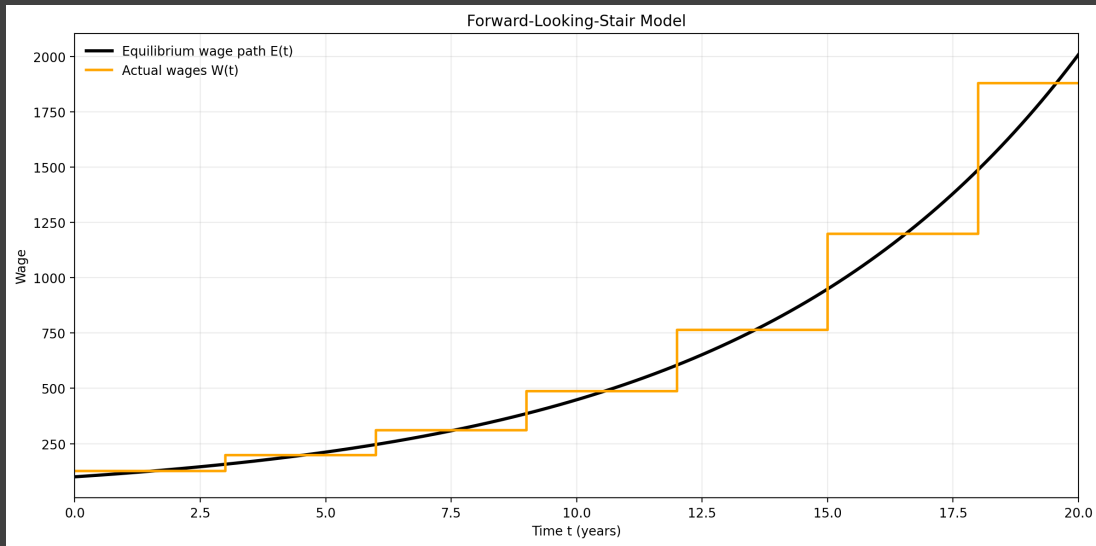


Figure 5. Forward-Looking Stair Model.

The latter is what happens when wage earners maximize expected utility given adequate information about inflation, and are financially literate enough to understand the implications. However wages could theoretically follow the former option, the lagging/backward-looking-stair model, without leading to persistently decreasing wage-share, since:

$$t_0 := \text{initial time}, \quad A := E(t_0), \quad r := \text{inflation rate}, \\ \Delta := \text{fixed wage-adjustment interval}$$

Let $E(t)$ denote the average, equilibrium labor market wage level in a frictionless price world:

$$E(t) = Ae^{r(t-t_0)}, \quad A > 0, \quad r > 0.$$

Suppose actual average wages update at fixed intervals of length $\Delta > 0$. For

$$t \in [t_0 + n\Delta, t_0 + (n+1)\Delta),$$

define actual average wages as

$$W(t) = E(t_0 + n\Delta) = Ae^{rn\Delta}.$$

Then

$$\frac{E(t)}{W(t)} = \frac{Ae^{r(t-t_0)}}{Ae^{rn\Delta}} = e^{r(t-t_0-n\Delta)}.$$

Since

$$0 \leq t - t_0 - n\Delta < \Delta,$$

we have

$$1 \leq e^{r(t-t_0-n\Delta)} < e^{r\Delta}.$$

Therefore,

$$1 \leq \frac{E(t)}{W(t)} < e^{r\Delta}.$$

Hence, $\frac{E(t)}{W(t)}$ is bounded above by the constant $e^{r\Delta}$ and bounded below by 1.

If the time between wage re-negotiations stays constant, and if wages followed the bounded-lagging-stair model, then you would expect wage-share to be lower in higher inflation environments (the area of discrepancy over time between frictionless price world equilibrium vs actual wage level would be larger), which is not shown to be the case (Malanga, 1974, pp. 8–9). However one could plausibly hypothesize that during high inflation environments, as inflation becomes a more salient issue to the wage-earner, periods between re-negotiations could systematically shorten compared to low inflation environments, and/or temporarily shift to a forward-looking-stair regime, both of which would keep wage-share constant or even increase it temporarily in high inflation environments, without invalidating the bounded-lagging-stair model during low/steady inflation.

Thus the bounded-lagging-stair model is plausible, and there is some limited empirical evidence pointing in that direction. [Hajdini et al. \(2022\)](#) suggests there is low pass through from inflation expectations to wage increase expectations on the part of wage earners. It is plausible that if wage earners do not expect their wages to keep up with inflation, then that suggests they will not bargain for it as intensely as they should if they were being rational, engaging in negotiation only when it becomes obvious to them they must to keep up with past inflation, without taking into account future inflation. The internal decision making process when deciding what a fair wage should be at renegotiation, for the average wage earner, may be based solely on a simple, implicit, intuitive, calculation/model informed by current price levels, without the capacity to account for predicted future changes in price levels. [Buchheim et al. \(2024\)](#), a limited study from Germany, find evidence of low passthrough from expected inflation to actual wage growth. A possible explanation could be that estimating future prices is more cognitively difficult, might require calculations and data, and is generally less intuitive. Another connected reason is that workers may be resistant to the marginal conflict of bargaining for a wage that feels too high at today's prices. Even though a wage earner would benefit from this same effect if it applied to other sticky contracts like loans, a lender is obviously far less likely to fall victim to this type of financial illiteracy / informational deficiency / cognitive barrier.

For a given business, the bounded-lagging-stair wage model would imply a persistent profit gap between equilibrium in an low/steady inflation economy, and the exact same business in a counterfactual fixed money supply economy, where nothing is changed but the rate of inflation. The result is an inflation tax-induced wealth redistribution from wage earners to equity owners. This effect is abstract, hard to measure, and hard to accept as a significant, structural, and persistent market inefficiency, but is still potentially worth exploring.

7 Conclusion

The ultimate downstream consequences I've described do not solely originate from a single wellspring of secular monetary expansion, but money printing *is*, I argue, upstream of all of them. The power to create money, the bedrock of markets, *out of nothing*, should not be taken lightly. It should be expected to have extremely broad and far reaching consequences. On the surface, the consequences of secular monetary growth seem stimulative, but we should consider this very carefully, since not all

growth is good. Some is eventually deadly, and inflation tax is uncannily similar to a malignant tumor: an unnatural corruption of the economic body's informational core, its DNA, an excessive growth, spreading slowly but unceasingly, causing harmful distortions in healthy processes, disguised as perfectly normal growth of the same fundamental building block as the rest of the system. As it currently stands, it seems we may be in desperate need of radical economic chemotherapy, a return to natural growth under hard money . . .

*“An imbalance between rich and poor is the oldest and most fatal ailment
of all republics.”*

— Plutarch

References

- Alchian, Armen A. and Reuben A. Kessel**, “The Meaning and Validity of the Inflation-Induced Lag of Wages Behind Prices,” *The American Economic Review*, 1960, 50 (1), 43–66.
- Alfageme, Ayoze**, “The Rise of Merger and Acquisitions in the US: Consequences for Investment, Market Concentration, and Profits—An Integrated and a Macroeconomic Approach with Firm-Level Data,” Working Paper 2505, Post-Keynesian Economics Society February 2025.
- Anadu, Kenechukwu, Mathias Kruttli, Patrick McCabe, Emilio Osambela, and Chae Hee Shin**, “The Shift from Active to Passive Investing: Potential Risks to Financial Stability?,” Finance and Economics Discussion Series 2018-060, Board of Governors of the Federal Reserve System August 2018.
- Asian Development Bank**, “Loan to CFPA Microfinance Management for Microfinance in Poverty-Stricken Counties: Sector Overview,” Report and Recommendation of the President RRP PRC 49432, Asian Development Bank 2016.
- Audretsch, David B. and Roy Thurik**, “Linking Entrepreneurship to Growth,” OECD Science, Technology and Industry Working Paper 2001/02, Organisation for Economic Co-operation and Development May 2001.
- Ayash, Brian and Mahdi Rastad**, “Leveraged Buyouts and Financial Distress,” *SSRN Electronic Journal*, October 2019.
- Ayyagari, Meghana, Asli Demirguc-Kunt, and Vojislav Maksimovic**, “Small vs. Young Firms across the World: Contribution to Employment, Job Creation, and Growth,” Policy Research Working Paper 5631, The World Bank April 2011.
- Bach, G. L.**, “Inflation: Who Gains and Who Loses?,” *Challenge*, 1974, 17 (3), 48–55.
- Banerjee, Ryan and Boris Hofmann**, “The Rise of Zombie Firms: Causes and Consequences,” *BIS Quarterly Review*, September 2018, pp. 67–78.
- Bank for International Settlements**, “Real Residential Property Prices for United States [QUSR628BIS],” FRED, Federal Reserve Bank of St. Louis 2026.
- Baumol, W. J. and W. G. Bowen**, “On the Performing Arts: The Anatomy of Their Economic Problems,” *The American Economic Review*, 1965, 55 (1/2), 495–502.
- Becker Friedman Institute for Economics**, “Direct Lenders Make Inroads into Middle Market Banking, but the Market Still Belongs to Traditional Providers,” Research Brief January 2026.

Berger, Allen N., Nathan H. Miller, Mitchell A. Petersen, Raghuram G. Rajan, and Jeremy C. Stein, “Does Function Follow Organizational Form? Evidence from the Lending Practices of Large and Small Banks,” NBER Working Paper 8752, National Bureau of Economic Research February 2002.

Berrospide, Jose, Fang Cai, Siddhartha Lewis-Hayre, and Filip Zikes, “Bank Lending to Private Credit: Size, Characteristics, and Financial Stability Implications,” FEDS Notes, Board of Governors of the Federal Reserve System May 2025.

Bhardwaj, Abhishek, Abhinav Gupta, and Sabrina T. Howell, “Capital Structure and Firm Outcomes: Evidence from Dividend Recapitalizations in Private Equity,” NBER Working Paper 33435, National Bureau of Economic Research January 2025.

Board of Governors of the Federal Reserve System, “Federal Reserve Bulletin, February Issues, 1948–1970,” Federal Reserve Bulletin 1948–1970. Monthly issues for February of each year, digitized by FRASER, Federal Reserve Bank of St. Louis.

—, “Annual Statistical Digest and Statistical Digest, 1970–2000,” Annual Statistical Digest / Statistical Digest 1981–2002. Includes Annual Statistical Digest 1970–1979, Annual Statistical Digest 1980–1989, Annual Statistical Digest 1990–1995, and Statistical Digest 1996–2000; digitized by FRASER, Federal Reserve Bank of St. Louis.

—, “Private Depository Institutions; Checkable and Time and Savings Deposits; Liability, Level (DISCONTINUED) [BOGZ1FL703139005Q],” FRED, Federal Reserve Bank of St. Louis 2024.

—, “Monetary Authority; Currency Outside Banks; Liability, Level,” FRED, Federal Reserve Bank of St. Louis 2025. Series ID: BOGZ1FL713125005A.

—, “Private Depository Institutions; Checkable Deposits; Liability, Level,” FRED, Federal Reserve Bank of St. Louis 2025. Series ID: BOGZ1FL703127005A.

—, “Private Depository Institutions; Total Time and Savings Deposits; Liability, Level,” FRED, Federal Reserve Bank of St. Louis 2025. Series ID: BOGZ1FL703130005A.

—, “DFA: Distributional Financial Accounts: Compare Wealth Components across Groups,” Board of Governors of the Federal Reserve System 2026.

—, “Federal Funds Effective Rate [FEDFUNDS],” FRED, Federal Reserve Bank of St. Louis 2026.

—, “Households; Checkable Deposits and Currency; Asset, Level [BOGZ1FL193020005Q],” FRED, Federal Reserve Bank of St. Louis 2026.

- , “Households; Total Assets, Level [BOGZ1FL192000005A],” FRED, Federal Reserve Bank of St. Louis 2026.
- , “Private Depository Institutions; Checkable Deposits; Liability, Level [BOGZ1FL703127005Q],” FRED, Federal Reserve Bank of St. Louis 2026.
- , “Private Depository Institutions; Total Time and Savings Deposits; Liability, Level [BOGZ1FL703130005Q],” FRED, Federal Reserve Bank of St. Louis 2026.
- , “Share of Checkable Deposits and Currency Held by the 50th to 90th Wealth Percentiles (DISCONTINUED) [WFRBSN40167],” FRED, Federal Reserve Bank of St. Louis 2026.
- , “Share of Checkable Deposits and Currency Held by the 90th to 99th Wealth Percentiles (DISCONTINUED) [WFRBSN09140],” FRED, Federal Reserve Bank of St. Louis 2026.
- , “Share of Checkable Deposits and Currency Held by the Bottom 50% (1st to 50th Wealth Percentiles) (DISCONTINUED) [WFRBSB50194],” FRED, Federal Reserve Bank of St. Louis 2026.
- , “Share of Checkable Deposits and Currency Held by the Top 1% (99th to 100th Wealth Percentiles) (DISCONTINUED) [WFRBST01113],” FRED, Federal Reserve Bank of St. Louis 2026.
- , “Share of Total Assets Held by the 50th to 90th Wealth Percentiles [WFRBSN40162],” FRED, Federal Reserve Bank of St. Louis 2026.
- , “Share of Total Assets Held by the 90th to 99th Wealth Percentiles [WFRBSN09135],” FRED, Federal Reserve Bank of St. Louis 2026.
- , “Share of Total Assets Held by the Bottom 50% (1st to 50th Wealth Percentiles) [WFRBSB50189],” FRED, Federal Reserve Bank of St. Louis 2026.
- , “Share of Total Assets Held by the Top 1% (99th to 100th Wealth Percentiles) [WFRBST01108],” FRED, Federal Reserve Bank of St. Louis 2026.

Borio, Claudio, Leonardo Gambacorta, and Boris Hofmann, “The Influence of Monetary Policy on Bank Profitability,” BIS Working Paper 514, Bank for International Settlements October 2015.

Buchak, Greg, Gregor Matvos, Tomasz Piskorski, and Amit Seru, “The Secular Decline of Bank Balance Sheet Lending,” NBER Working Paper 32176, National Bureau of Economic Research October 2024. Originally issued February 2024; revised October 2024.

Buchheim, Lukas, Sebastian Link, and Sascha Möhrle, “Inflation and Wage Expectations of Firms and Employees,” Working Paper, TU Dortmund and ifo Institute and University of Munich August 2024.

Christophers, Brett, “Financialisation as Monopoly Profit: The Case of US Banking,” *Antipode*, 2018.

Congressional Research Service, “Productivity Growth: Trends and Policy Issues,” CRS Report R48695 September 2025.

Davydiuk, Tetiana, Tatyana Marchuk, and Samuel Rosen, “Direct Lenders in the U.S. Middle Market,” Technical Report, Working paper December 2020.

Decker, Ryan A. and John Haltiwanger, “Surging Business Formation in the Pandemic: A Brief Update,” Technical Report, Federal Reserve Board September 2024.

—, —, **Ron S. Jarmin, and Javier Miranda**, “Declining Business Dynamism: What We Know and the Way Forward,” *American Economic Review: Papers & Proceedings*, 2016, 106 (5), 203–207.

Duan, Kun, Plamen Ivanov, and Richard Werner, “Deciphering the Chinese Economic Miracle: The Resolution of an Age-Old Economists’ Debate—and its Central Role in Rapid Economic Development,” *Review of Political Economy*, April 2023.

Easterly, William and Stanley Fischer, “Inflation and the Poor,” *Journal of Money, Credit and Banking*, 2001, 33 (2), 160–178.

Eiteman, Wilford J., “The Regulation of Brokers’ Loans,” *The American Economic Review*, 1933, 23 (3), 456–465.

Erel, Isil and Eduard Inozemtsev, “Evolution of Debt Financing Toward Less-Regulated Financial Intermediaries in the United States,” *Journal of Financial and Quantitative Analysis*, 2025, 60 (3), 1234–1271.

Favara, Giovanni, Ivan Ivanov, and Marcelo Rezende, “GSIB Surcharges and Bank Lending: Evidence from US Corporate Loan Data,” *Journal of Financial Economics*, 2021, 142 (3), 1426–1443.

Federal Deposit Insurance Corporation, “Capital Maintenance; Final Statement of Policy on Risk-Based Capital,” *Federal Register*, March 1989, 54 (53), 11500–11517.

—, “FDIC Community Banking Study,” Technical Report, Federal Deposit Insurance Corporation December 2012.

—, “Risk Review 2025,” Technical Report, Federal Deposit Insurance Corporation 2025.

Federal Reserve Bank of St. Louis, “Release Tables: Shares of Wealth by Wealth Percentile Groups,” FRED, Distributional Financial Accounts 2025.

- Feinman, Joshua N.**, “Reserve Requirements: History, Current Practice, and Potential Reform,” *Federal Reserve Bulletin*, June 1993, pp. 569–589.
- Financial Industry Regulatory Authority**, “Customer Margin Balances,” Margin Statistics, Excel spreadsheet 2026. Data through March 2026; includes debit balances in customers’ securities margin accounts, free credit balances in customers’ cash accounts, and free credit balances in customers’ securities margin accounts.
- Fritzdixon, Kathryn**, “Bank and Nonbank Lending Over the Past 70 Years,” *FDIC Quarterly*, 2019, 13 (4), 31–39.
- Gopal, Manasa and Philipp Schnabl**, “The Rise of Finance Companies and FinTech Lenders in Small Business Lending,” Technical Report, Working paper September 2020.
- Gordon, Robert J.**, “The Demise of U.S. Economic Growth: Restatement, Rebuttal, and Reflections,” Technical Report, Northwestern University and National Bureau of Economic Research January 2014.
- Green, Michael W.**, “Part 2: The Door Has Opened,” Yes, I Give a Fig November 2025.
- Gupta, Atul, Sabrina T. Howell, Constantine Yannelis, and Abhinav Gupta**, “Owner Incentives and Performance in Healthcare: Private Equity Investment in Nursing Homes,” *The Review of Financial Studies*, 2024, 37 (4), 1029–1077.
- Hajdini, Ina, Edward S. Knotek II, John Leer, Mathieu Pedemonte, Robert Rich, and Raphael Schoenle**, “Low Passthrough from Inflation Expectations to Income Growth Expectations: Why People Dislike Inflation,” Working Paper, Federal Reserve Bank of Cleveland and Brandeis University and Morning Consult September 2022.
- Hamel, Gary and Michele Zanini**, “The \$3 Trillion Prize for Busting Bureaucracy: And How to Claim It,” Technical Report, London Business School and The Management Lab March 2016.
- Hanson, Samuel G., Victoria Ivashina, Laura Nicolae, Jeremy C. Stein, Adi Sunderam, and Daniel K. Tarullo**, “The Evolution of Banking in the 21st Century: Evidence and Regulatory Implications,” March 2024. Brookings Papers on Economic Activity Conference Draft, Spring 2024.
- Hoffman, Michael**, “Insured Institution Performance,” *FDIC Quarterly*, 2025, 19 (1).
- Jordà, Òscar, Moritz Schularick, and Alan M. Taylor**, “The Great Mortgaging: Housing Finance, Crises, and Business Cycles,” Working Paper 2014-23, Federal Reserve Bank of San Francisco September 2014.

- Kahn, R. Jay and Luke M. Olson**, “Who Participates in Cleared Repo?,” Office of Financial Research Brief Series July 2021.
- Kannan, Sneha, Joseph Dov Bruch, and Zirui Song**, “Changes in Hospital Adverse Events and Patient Outcomes Associated With Private Equity Acquisition,” *JAMA*, 2023, *330* (24), 2365–2375.
- Knoll, Katharina, Moritz Schularick, and Thomas Steger**, “No Price Like Home: Global House Prices, 1870–2012,” *American Economic Review*, 2017, *107* (2), 331–353.
- Kofman, Daniel**, “Reconsidering Money and Inflation: Groundwork for an Alternative Macro Basis,” May 2026. Unpublished manuscript.
- Kopp, Emanuel and Peter D. Williams**, “A Macroeconomic Approach to the Term Premium,” IMF Working Paper WP/18/140, International Monetary Fund June 2018.
- Maharrey, Mike**, “For First Time in Nearly 30 Years, Foreign Central Banks Hold More Gold Than U.S. Treasuries,” FXStreet August 2025.
- Malanga, Vincent**, “The Impact of Inflation on the Distribution of Income: A Survey,” Research Paper 7417, Federal Reserve Bank of New York September 1974.
- Martin-Buck, Frank**, “Leveraged Lending and Corporate Borrowing: Increased Reliance on Capital Markets, With Important Bank Links,” *FDIC Quarterly*, 2019, *13* (4), 41–50.
- Mauboussin, Michael J.**, “Surge in the Urge to Merge: M&A: Trends and Analysis,” Legg Mason Capital Management January 2010.
- McCord, Roisin and Edward Simpson Prescott**, “The Financial Crisis, the Collapse of Bank Entry, and Changes in the Size Distribution of Banks,” *Federal Reserve Bank of Richmond Economic Quarterly*, 2014, *100* (1), 23–50.
- Mian, Atif, Ludwig Straub, and Amir Sufi**, “The Saving Glut of the Rich,” Technical Report, Princeton University, Harvard University, and University of Chicago Booth School of Business July 2025. Working paper.
- Mkhaiber, Achraf and Richard A. Werner**, “The Relationship between Bank Size and the Propensity to Lend to Small Firms: New Empirical Evidence from a Large Sample,” *Journal of International Money and Finance*, 2021, *110*, 102281.
- multpl.com**, “Shiller PE Ratio,” multpl.com 2026.
- National Bureau of Economic Research**, “Total Deposits, All Commercial Banks for United States [M1445AUSM144SNBR],” FRED, Federal Reserve Bank of St. Louis 2026.

- Obernberger, Stefan, Stefan Weik, and Xiang Yin**, “Private Equity and the Organization of Firms,” SSRN January 2026.
- Press, Robert**, “Unpacking the Declining Receipts Share of Small Firms,” Issue Brief 20, U.S. Small Business Administration, Office of Advocacy May 2024.
- Rajan, Raghuram and Rodney Ramcharan**, “The Anatomy of a Credit Crisis: The Boom and Bust in Farm Land Prices in the United States in the 1920s,” *American Economic Review*, 2015, 105 (4), 1439–1477.
- Rappoport, Peter and Eugene N. White**, “Was There a Bubble in the 1929 Stock Market?,” NBER Working Paper 3612, National Bureau of Economic Research, Cambridge, MA February 1991.
- Senior Supervisors Group**, “Risk Management Lessons from the Global Banking Crisis of 2008,” Technical Report, Senior Supervisors Group October 2009.
- Shiller, Robert J.**, “Why Do People Dislike Inflation?,” Cowles Foundation Discussion Paper 1115, Cowles Foundation for Research in Economics, Yale University March 1996.
- Stern, Gary H. and Ron J. Feldman**, *Too Big to Fail: The Hazards of Bank Bailouts*, Washington, DC: Brookings Institution Press, 2004.
- United States Senate Committee on Banking and Currency**, “Stock Exchange Practices: Hearings Before the Committee on Banking and Currency, United States Senate, Seventy-Third Congress, First Session, Part 15: National Securities Exchange Act 1934,” United States Government Printing Office 1934.
- Werner, Richard A.**, “Towards a New Monetary Paradigm: A Quantity Theorem of Disaggregated Credit, with Evidence from Japan,” *Kredit und Kapital*, 1997, 30 (2), 276–309.
- Wheelock, David C.**, “Regulation, Market Structure, and the Bank Failures of the Great Depression,” *Federal Reserve Bank of St. Louis Review*, March/April 1995, pp. 27–38.
- Zook, Chris and James Allen**, “Barriers and Pathways to Sustainable Growth: Harnessing the Power of the Founder’s Mentality,” Technical Report, Bain & Company 2016.